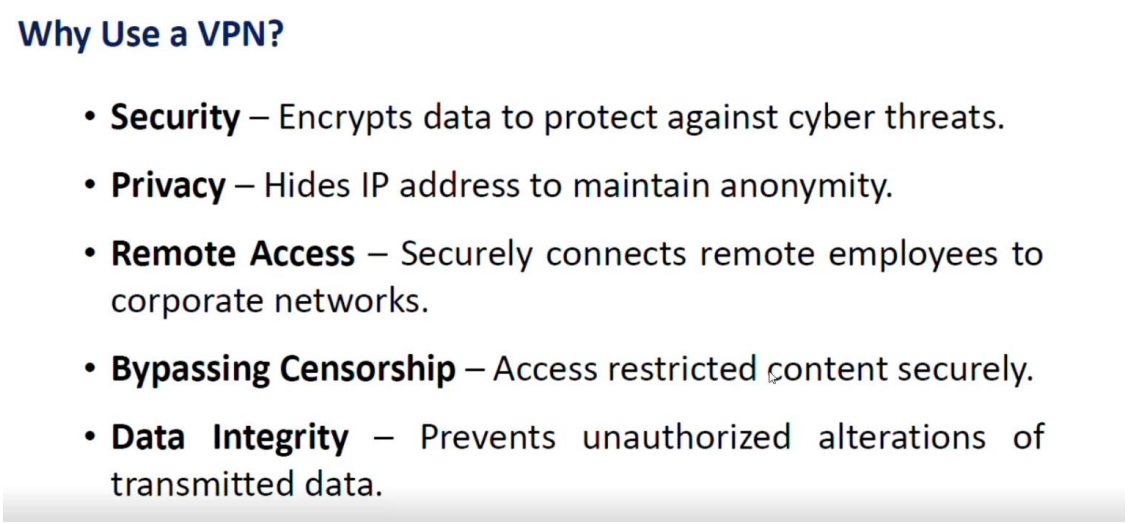
installation. Once it comes back up, the RDS deployment exists but still needs licensing (Part 2) before it can be used beyond the two-session admin grace period.

Part 2 — Virtual Private Network (VPN): Concepts

This section (marked "12th Day" in the original notes) is pure theory covering what a VPN is and why it matters, before the guide returns to finishing the RDS configuration (RD Licensing and RD Gateway) and then to actually building a VPN server with RRAS. The theory is included here in full because it explains the reasoning behind several configuration choices made later (e.g. why VPN Access is enabled but Dial-up Access is not, or why NAT is combined with VPN).

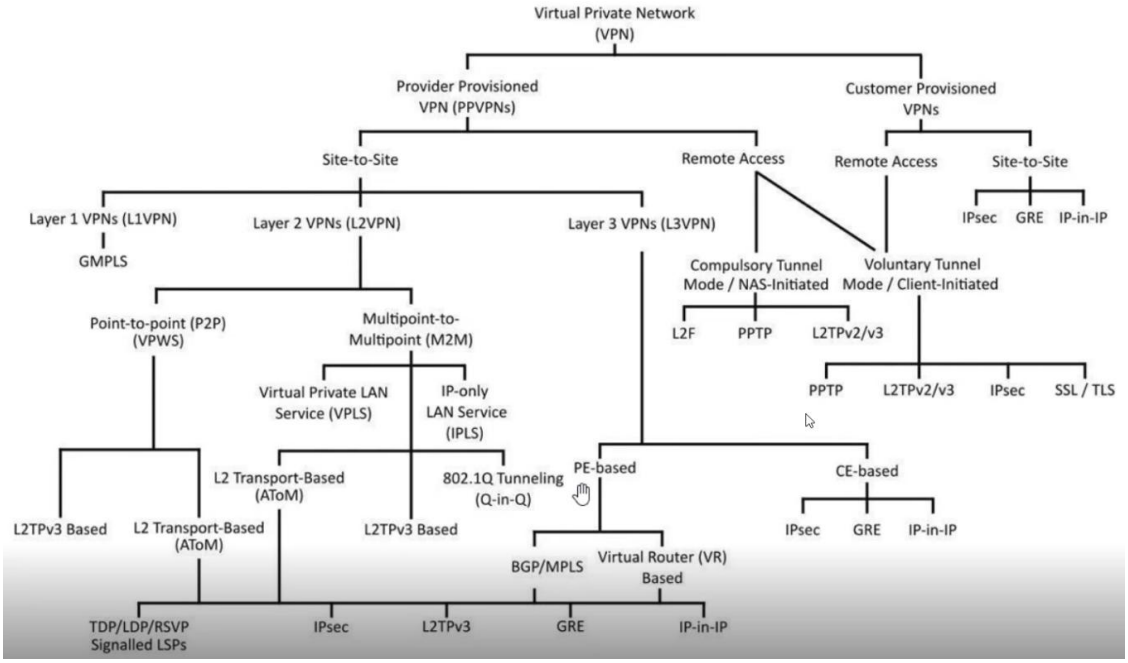
2.1 Why use a VPN?



What this screen shows: A summary slide listing five reasons organisations use VPNs: Security (encrypts data against cyber threats), Privacy (hides the user's real IP address), Remote Access (securely connects remote employees to the corporate network — this is the use case this guide builds), Bypassing Censorship (accessing restricted content securely), and Data Integrity (preventing unauthorised alteration of data in transit).

What happens next: For this guide, the relevant reason is Remote Access: employees working outside the office need a secure tunnel back into the nextgen.int network so they can reach internal resources such as file shares and the RDS deployment built in Part 1.

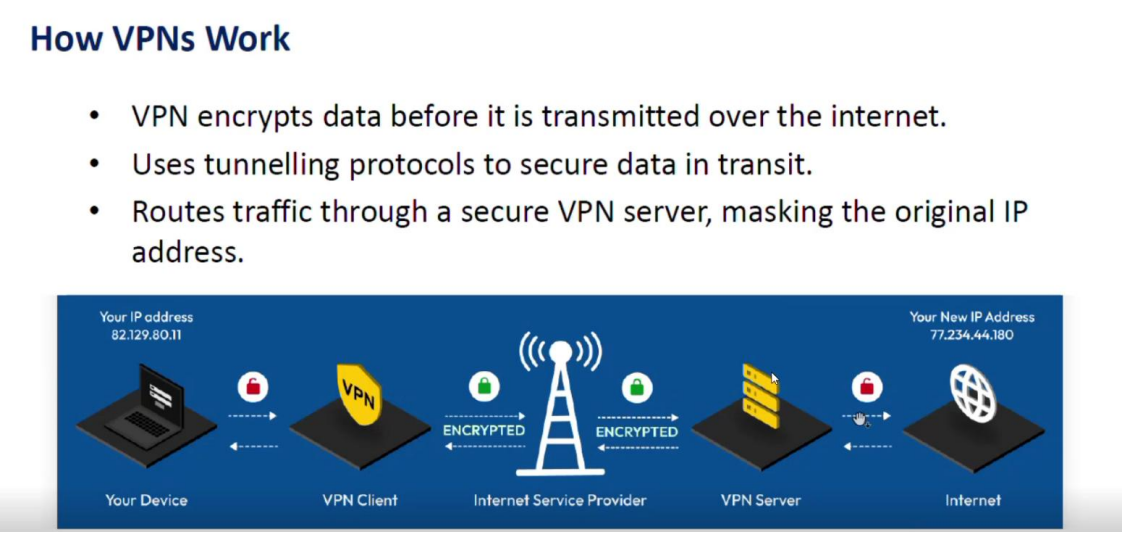
2.2 VPN taxonomy



What this screen shows: A large classification diagram of VPN types. At the top level VPNs split into Provider Provisioned VPNs (PPVPNs) and Customer Provisioned VPNs. Provider-provisioned VPNs branch into Site-to-Site and Remote Access categories, further divided by OSI layer (Layer 1/2/3 VPNs) and by technology (MPLS, VPLS, L2TPv3, BGP/MPLS, etc.). Customer-provisioned VPNs branch into Remote Access (Compulsory Tunnel / Voluntary Tunnel, using protocols such as L2F, PPTP, L2TPv2/v3, IPsec, PPTP, SSL/TLS) and Site-to-Site (IPsec, GRE, IP-in-IP).

What happens next: This diagram is mainly for context: it shows that "VPN" is an umbrella term covering many different technologies. The specific technology this guide implements later — Windows RRAS with a VPN Gateway offering SSTP/L2TP/PPTP/IKEv2 — sits in the Customer Provisioned → Remote Access → Voluntary Tunnel branch of this tree.

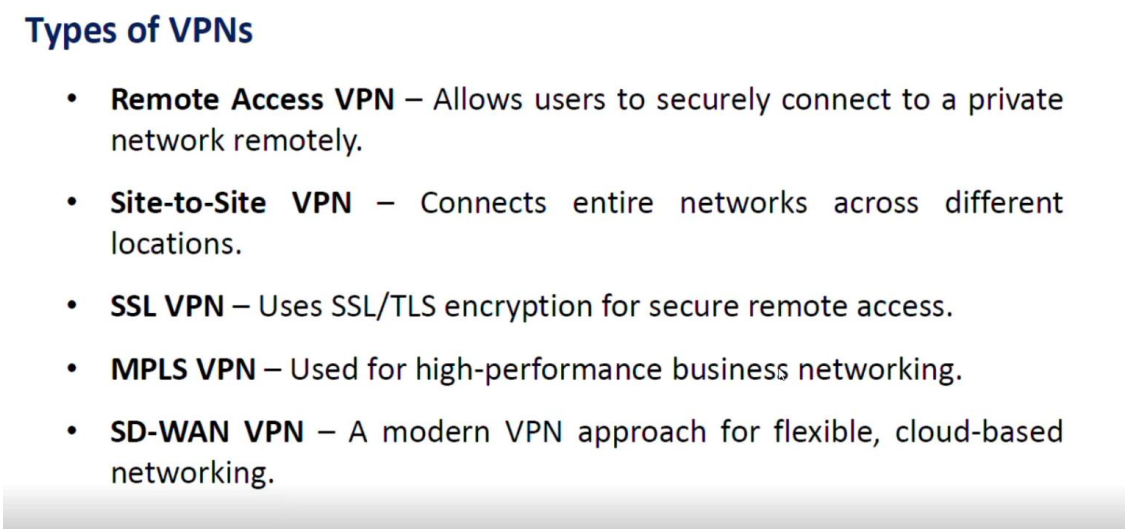
2.3 How VPNs work



What this screen shows: A diagram showing the end-to-end path of a VPN connection: the user's device has its real IP address (shown as 82.129.00.31); the VPN client encrypts the traffic; it passes through the Internet Service Provider; the encrypted tunnel reaches the VPN server; from there it is decrypted and sent to the Internet, now appearing to originate from a new IP address (shown as 17.234.44.180) instead of the user's own.

What happens next: This is the mental model for everything built later: the client encrypts before leaving the device, the ISP only ever sees encrypted traffic it cannot read, and the VPN server is the only point where the traffic is decrypted and forwarded onward.

2.4 Types of VPN



What this screen shows: A slide listing five VPN categories: Remote Access VPN (an individual user connecting securely to a private network — the type this guide builds), Site-to-Site VPN (connecting entire networks/offices together), SSL VPN (uses SSL/TLS encryption for secure remote access, typically browser- or client-based), MPLS VPN (used for high-performance business networking between sites, usually provider-managed), and SD-WAN VPN (a modern, flexible, cloud-based networking approach).

What happens next: Confirms again that the configuration built in Part 4 of this guide is a Remote Access VPN: a single remote user's device connecting into the nextgen.int network, not a permanent site-to-site link between two offices.

2.5 VPN protocols

VPN Protocols

- **IPSec (Internet Protocol Security)** – Secure data at the network layer
- **SSL/TLS (Secure Sockets Layer/Transport Layer Security)** – Used in web-based VPNs
- **L2TP/IPSec (Layer 2 Tunneling Protocol)** – Provides encryption and authentication
- **OpenVPN** – Open-source, flexible, and secure
- **WireGuard** – Modern, faster VPN with enhanced security

What this screen shows: A list of the underlying protocols VPNs are built on: IPSec (secures data at the network layer), SSL/TLS (used in web-based VPNs), L2TP/IPSec (Layer 2 Tunneling Protocol, combined with IPSec for encryption and authentication), OpenVPN (open-source, flexible, secure), and WireGuard (a modern, faster VPN protocol with enhanced security).

What happens next: Relevant later: when the RRAS server is configured, it opens IP Security (IKE) on UDP port 500 — this is the IPSec/IKE protocol referenced here, used because Windows' built-in VPN Gateway supports IKEv2/IPSec as one of its tunnel types.

2.6 VPN security considerations

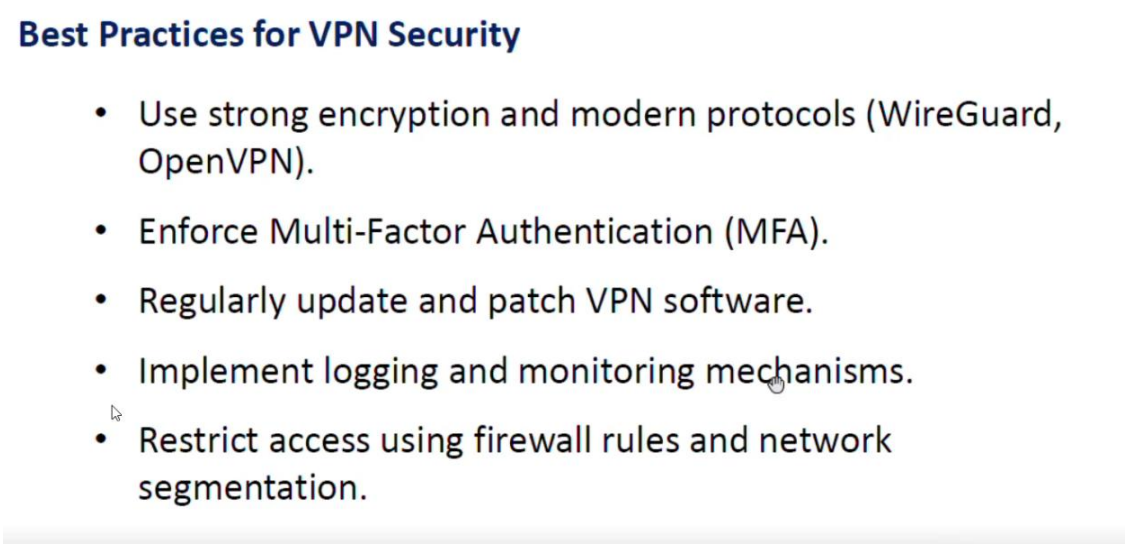
VPN Security Considerations

- **Encryption Standards** – AES-256, ChaCha20 for strong security.
- **Authentication Mechanisms** – Multi-Factor Authentication (MFA), digital certificates.
- **Firewall and Access Control** – Configuring firewalls to allow VPN traffic while blocking unauthorized access.
- **Logging and Monitoring** – Track VPN usage to detect potential security threats.

What this screen shows: A slide on hardening a VPN: Encryption Standards (AES-256, ChaCha20 for strong security), Authentication Mechanisms (Multi-Factor Authentication, digital certificates), Firewall and Access Control (configuring firewalls to allow VPN traffic while blocking unauthorised access), and Logging and Monitoring (tracking VPN usage to detect potential threats).

What happens next: This is why the guide later configures Network Policy Server (NPS) policies explicitly for the VPN connections, and why the RRAS server's firewall/NAT rules only open the specific ports the VPN protocols need (rather than leaving everything open).

2.7 Best practices for VPN security



What this screen shows: A checklist: use strong encryption and modern protocols (WireGuard, OpenVPN); enforce Multi-Factor Authentication (MFA); regularly update and patch VPN software; implement logging and monitoring mechanisms; restrict access using firewall rules and network segmentation.

What happens next: Worth treating as a production checklist to apply on top of the base configuration in this guide — the walkthrough itself sets up basic VPN access and NAT, but a real deployment should also add MFA and stricter firewall segmentation before going live.

2.8 Enterprise VPN solution examples



What this screen shows: The Work Resources page loaded successfully from the client, showing the same published icons — Calculator, Calculator, Notepad, Paint, Paint, XPS Viewer.

What to do here: Click any icon to test that a RemoteApp session actually launches end-to-end from an external/non-domain-joined client.

What happens next: This is the final proof-of-concept for Part 3: a client machine that is not domain-joined, with only the certificate imported and a valid license, can browse to RD Web Access over HTTPS and launch published applications through RD Gateway.

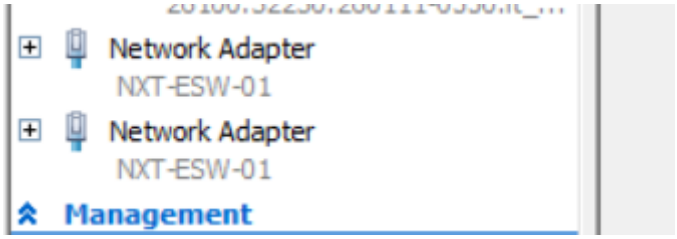
Reference architecture for this flow, from the notes: User PC → HTTPS (443) → RD Gateway → RDP (3389) → RDS Server. The client only ever speaks HTTPS to the Gateway; the Gateway is what relays the traffic internally over RDP to the Session Host, so RDP itself is never exposed to the internet.

Part 4 — Building the VPN Server with RRAS

This is the practical continuation of the VPN theory from Part 2. A Routing and Remote Access Service (RRAS) role is added to a server so that remote users can connect over the internet and reach internal resources through an encrypted tunnel, controlled by network policies and address pools configured on that server.

4.1 Preparing the network adapters

4.1.1 Adding a second network adapter

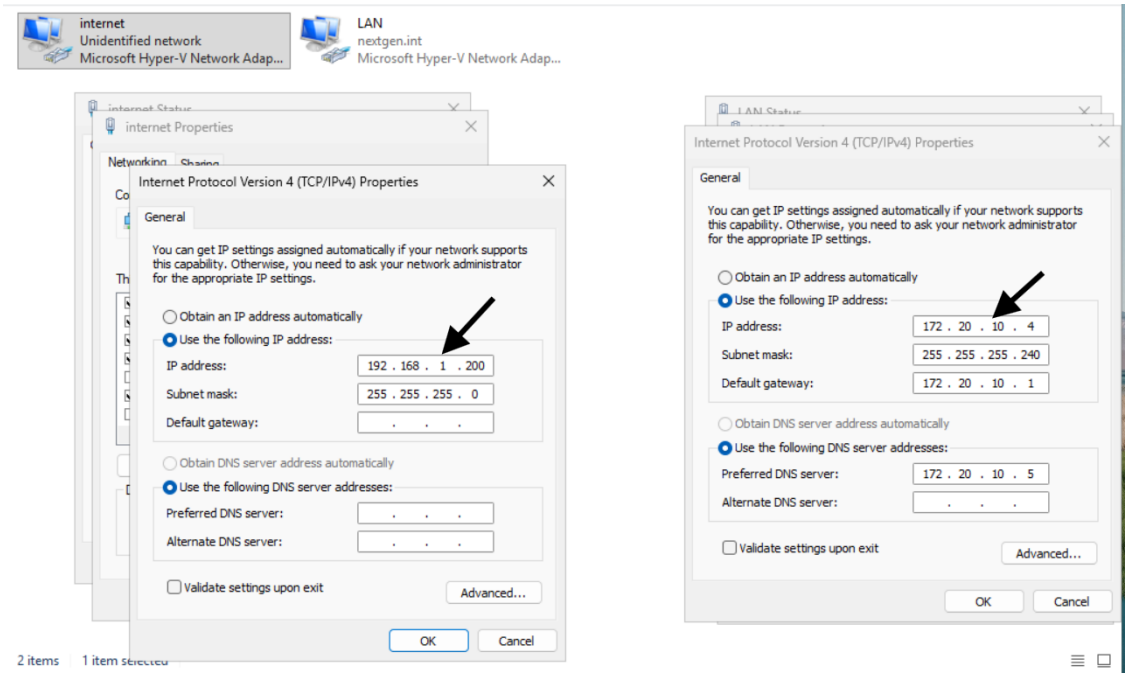


What this screen shows: Hyper-V virtual machine settings showing two Network Adapter entries both currently attached to the NXT-ESW-01 virtual switch, plus a Management section below.

What to do here: Add a second (additional) virtual network adapter to the AD/VPN server so it has two distinct network paths: one representing the internal/LAN side and one representing the internet-facing side. Attach each adapter to a different virtual switch so they sit on genuinely different networks — using the same switch/network for both defeats the purpose of a public/private interface split needed later for NAT.

What happens next: The server now has two NICs available to configure with static IPs in the next step — one will become the 'LAN' interface and the other the 'internet' interface referenced throughout the rest of this section.

4.1.2 Configuring static IPs on both adapters

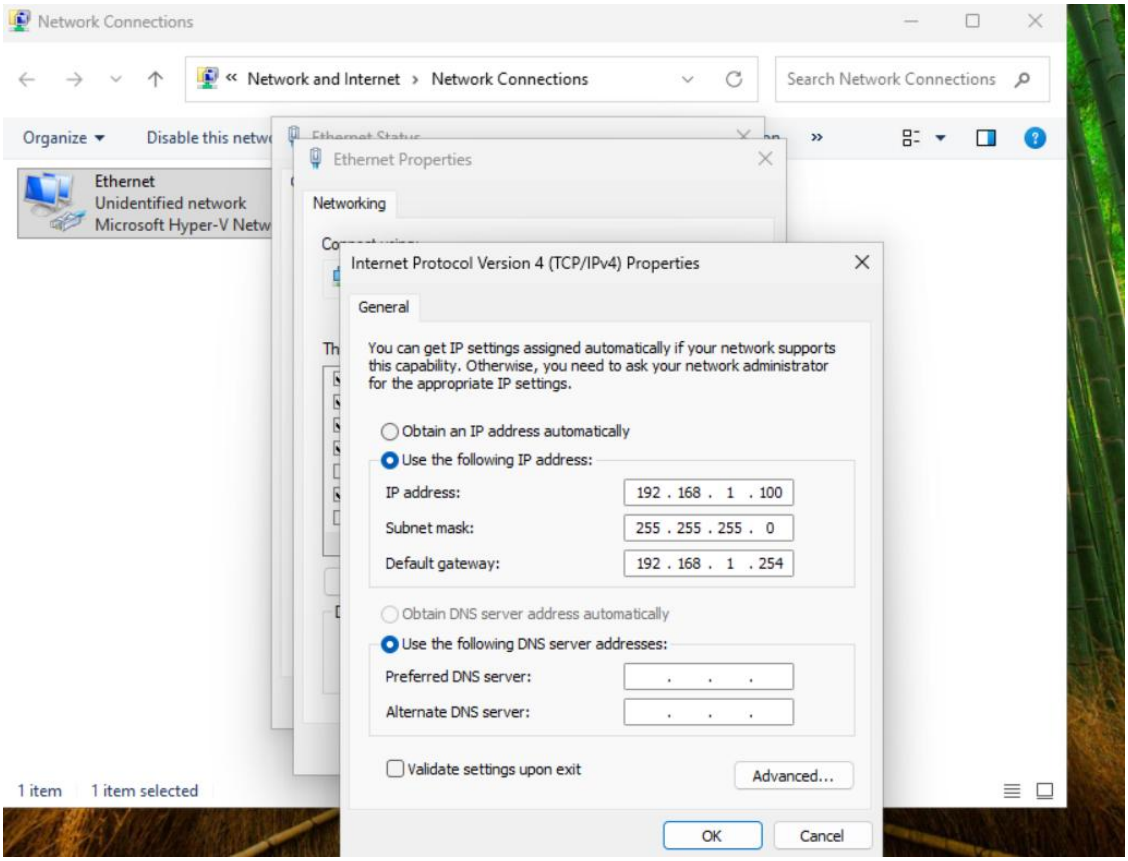


What this screen shows: Two side-by-side IPv4 Properties windows. The left one (labelled 'Nekhwinkm... Home' in the taskbar) sets IP address 192.168.1.200, Subnet mask 255.255.255.0, no gateway configured; the right one sets IP address 172.20.10.4, Subnet mask 255.255.255.240, Default gateway 172.20.10.1, with preferred DNS server 172.20.10.5 — both indicated by arrows pointing at the 'Use the following IP address' fields.

What to do here: On the internal-facing adapter, set a static IP in the internal network range (here 192.168.1.200/24) with no default gateway (since internet-bound traffic should go out the other adapter, not this one). On the internet-facing adapter, set a static IP matching whatever network provides internet access (here 172.20.10.4/28) along with its default gateway (172.20.10.1) and DNS server (172.20.10.5).

What happens next: Each adapter now has a fixed, predictable IP address — required because RRAS needs to know exactly which interface is 'internal' and which is 'internet' when it is configured a few steps later, and DHCP-assigned addresses could change and break that configuration.

4.1.3 Configuring the client PC's network adapter



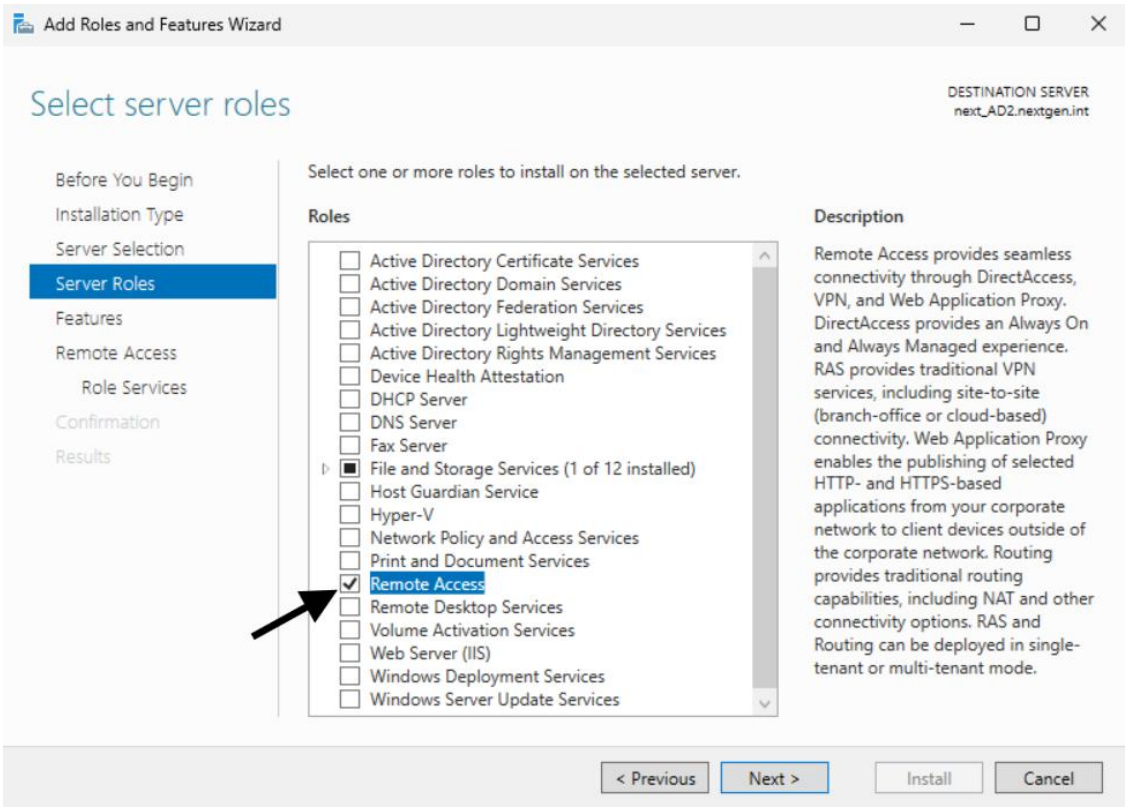
What this screen shows: The Network Connections control panel showing an Ethernet adapter, with its Ethernet Properties → Internet Protocol Version 4 (TCP/IPv4) Properties dialog open: IP address 192.168.1.100, Subnet mask 255.255.255.0, Default gateway 192.168.1.254; DNS servers left blank/automatic.

What to do here: On the client PC that will simulate a remote user connecting over the internet (this PC is deliberately not joined to the domain), configure it as if it were a computer on the internet-facing network the VPN server's public interface would be reachable from, with an IP address in the same subnet as the server's internet-facing interface plus the correct default gateway.

What happens next: This client is now positioned as an 'outside' machine relative to the AD network, so that connecting to the VPN later genuinely tests remote access rather than access that already works because the client sits on the internal LAN.

4.2 Installing the Remote Access (RRAS) role

4.2.1 Selecting the Remote Access server role

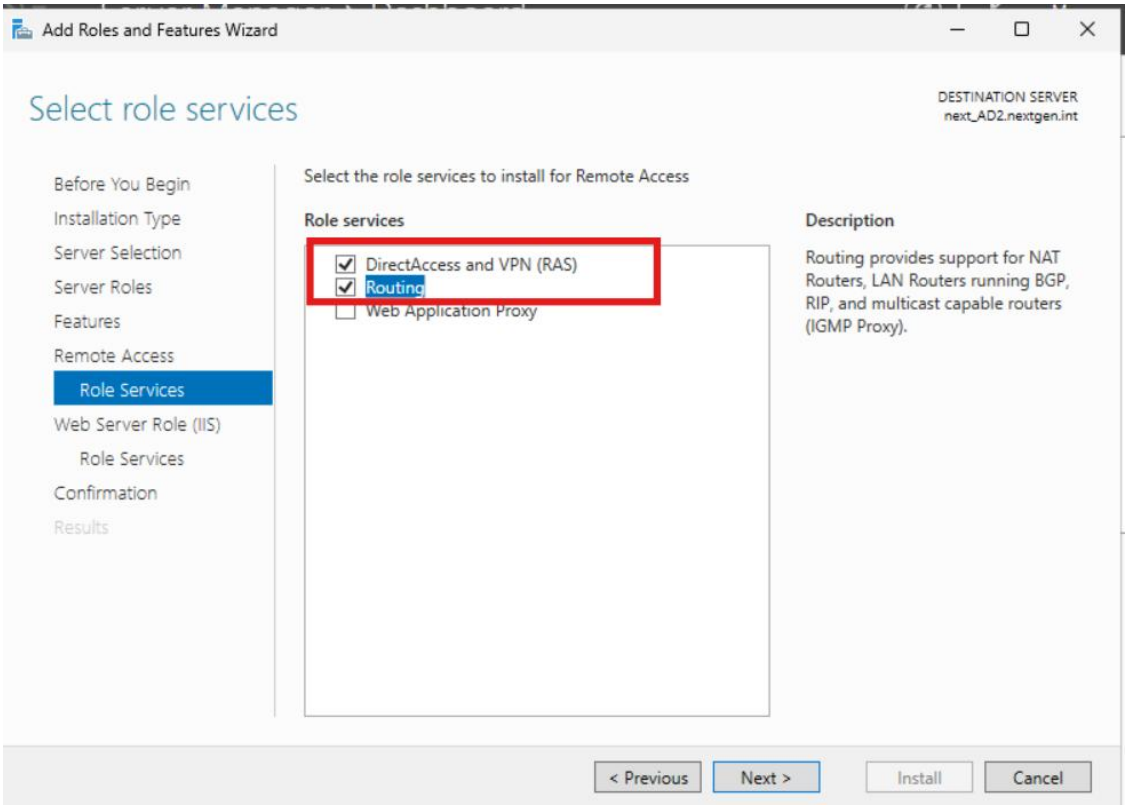


What this screen shows: The Add Roles and Features Wizard, Select server roles page, with Remote Access ticked (arrow pointing at it) in the Roles checklist alongside other roles like Active Directory Certificate Services, DHCP Server, DNS Server, Hyper-V, Remote Desktop Services (already installed), Web Server (IIS), etc. The description panel explains Remote Access provides seamless connectivity through DirectAccess, VPN, and Web Application Proxy; DirectAccess provides an Always On/Always Managed experience; RAS provides traditional VPN services including site-to-site (branch-office or cloud-based) connectivity.

What to do here: Tick Remote Access and click Next.

What happens next: The wizard proceeds to Role Services, where the specific Remote Access sub-features (DirectAccess/VPN vs Web Application Proxy) are chosen.

4.2.2 Selecting role services

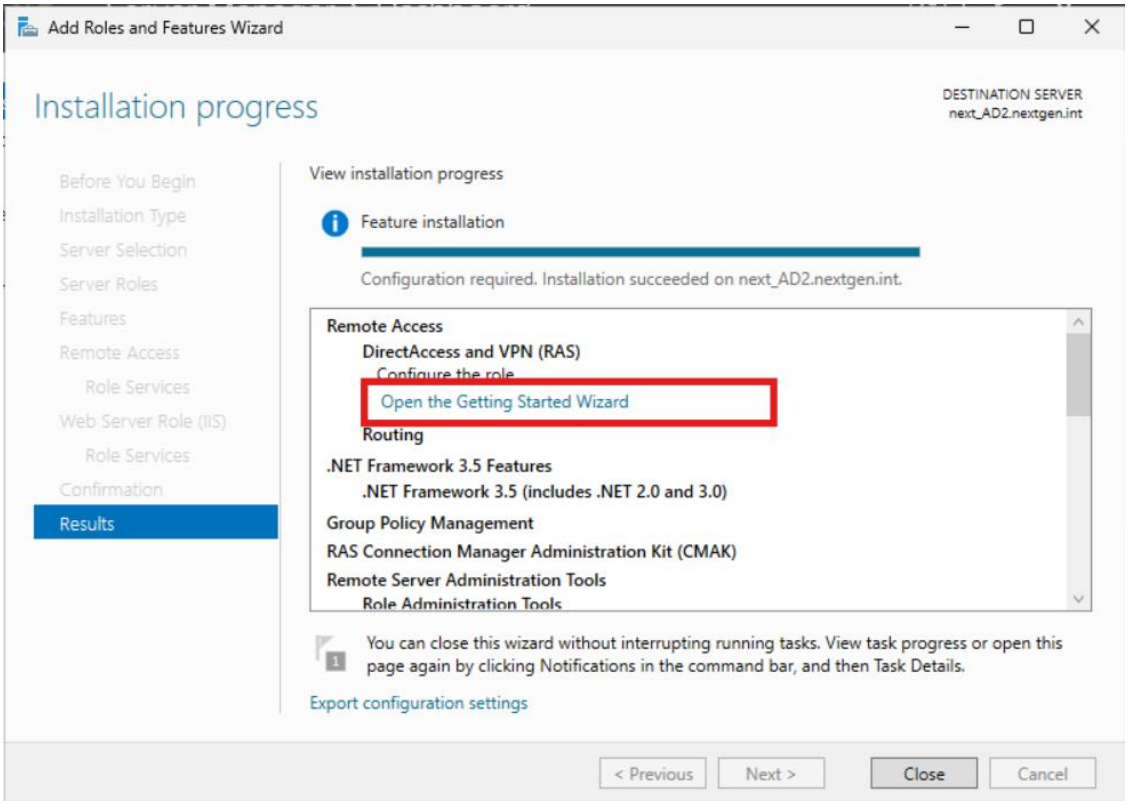


What this screen shows: The Select role services page under Remote Access, with two checkboxes ticked: DirectAccess and VPN (RAS) and Routing (both highlighted in a red box); Web Application Proxy is left unticked. The description explains Routing provides support for NAT Routers, LAN Routers running BGP, RIP, and multicast-capable routers (IGMP Proxy).

What to do here: Tick both DirectAccess and VPN (RAS) — this is the component that actually implements the VPN server — and Routing, since Routing is what enables the NAT functionality configured later so internal clients (and VPN clients) can reach the internet through this same server. Leave Web Application Proxy unticked, since that role is for reverse-proxying web apps, not relevant here.

What happens next: The wizard proceeds to install the RAS and Routing role services.

4.2.3 Installation progress and Getting Started Wizard

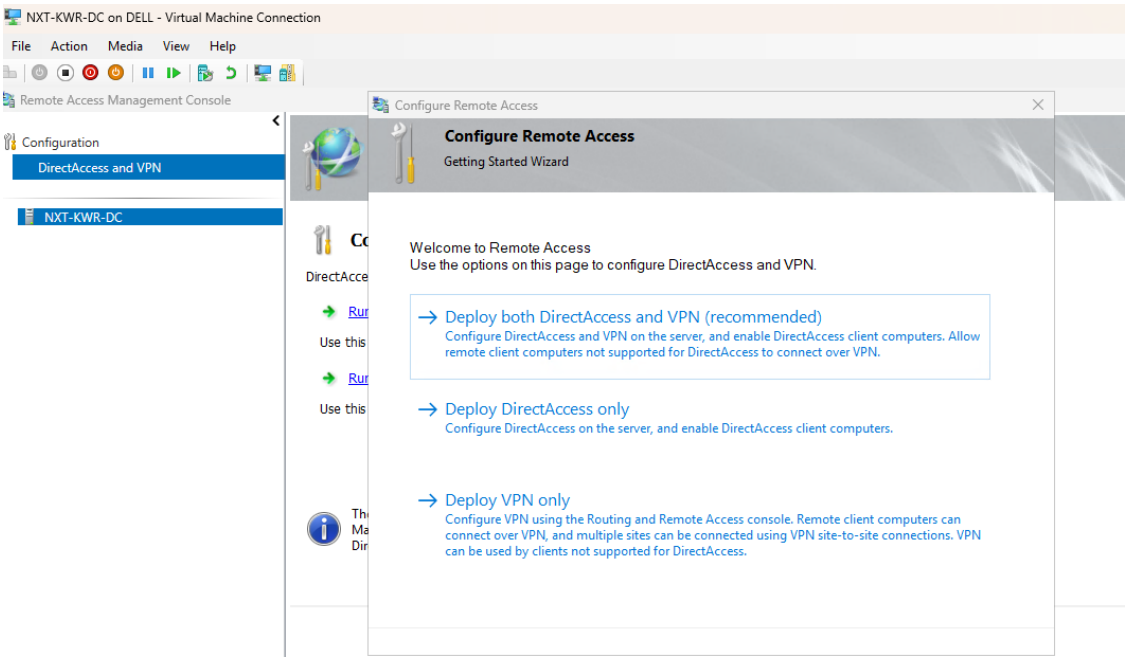


What this screen shows: The Installation progress page confirms 'Configuration required. Installation succeeded on next_AD2.nextgen.int', listing Remote Access → DirectAccess and VPN (RAS) → Configure the role → Open the Getting Started Wizard (highlighted in a red box), plus Routing, .NET Framework 3.5 Features, Group Policy Management, RAS Connection Manager Administration Kit (CMAK), Remote Server Administration Tools, and Role Administration Tools as supporting features that were installed alongside it.

What to do here: Click Open the Getting Started Wizard to move straight into configuring the newly installed Remote Access role (rather than leaving it installed-but-unconfigured).

What happens next: The Configure Remote Access wizard opens, offering the high-level deployment choice covered next.

4.2.4 Choosing a Remote Access deployment option

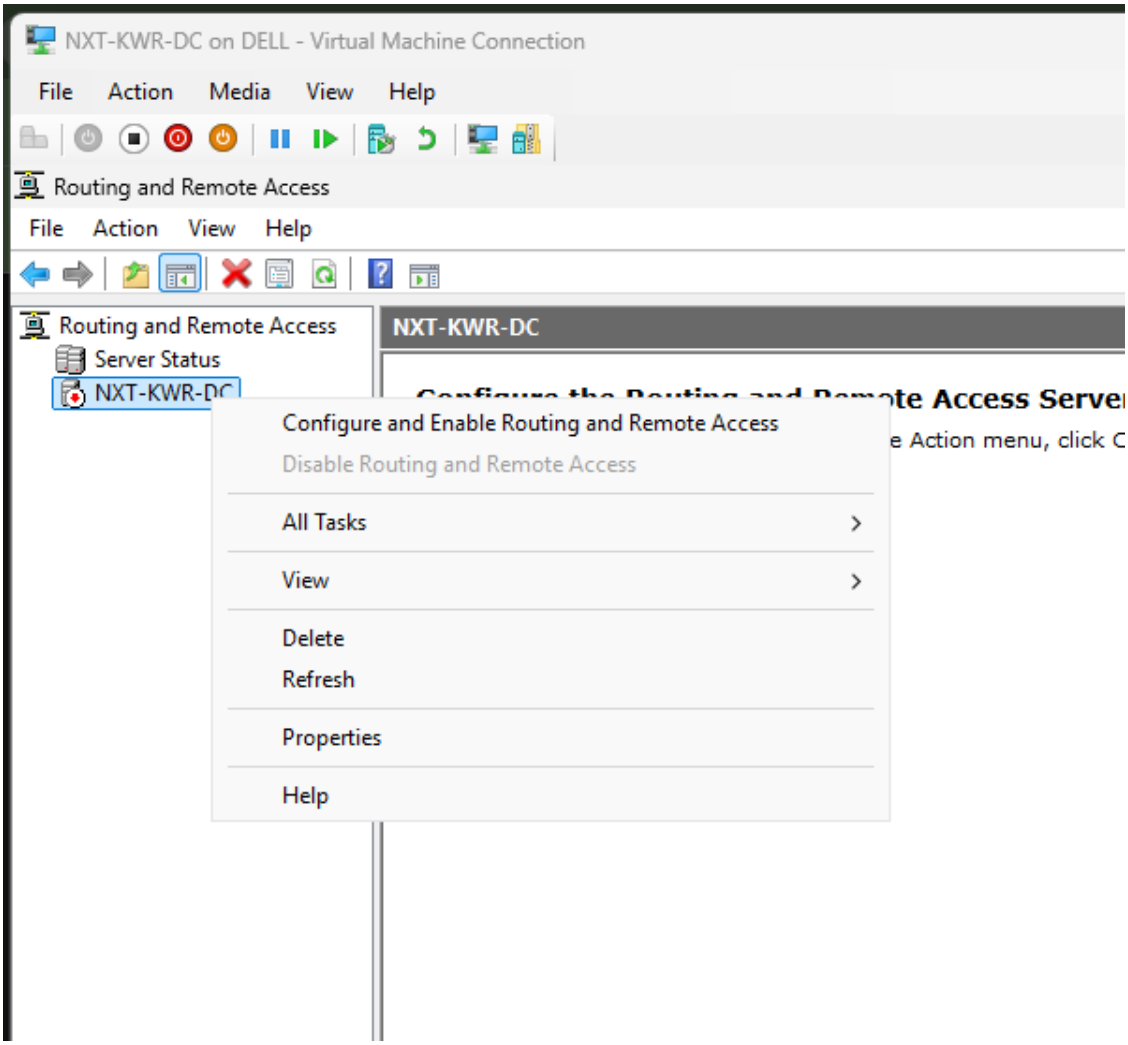


What this screen shows: The Configure Remote Access — Getting Started Wizard, offering three large clickable options: Deploy both DirectAccess and VPN (recommended) — configure DirectAccess and VPN on the server and enable DirectAccess client computers, plus allow remote client computers not supported for DirectAccess to connect over VPN; Deploy DirectAccess only; and Deploy VPN only — configure VPN using the Routing and Remote Access console, multiple sites can be connected using VPN site-to-site connections, and VPN can be used by clients not supported for DirectAccess.

What to do here: Click Deploy VPN only, since this guide's goal is a straightforward remote-access VPN rather than the more complex, certificate- and Group-Policy-heavy DirectAccess technology (DirectAccess also requires domain-joined, DirectAccess-capable Windows clients, which does not match the non-domain-joined client used for testing).

What happens next: This launches the classic Routing and Remote Access snap-in directly to its setup wizard instead of the newer DirectAccess configuration console.

4.2.5 Enabling Routing and Remote Access from the console

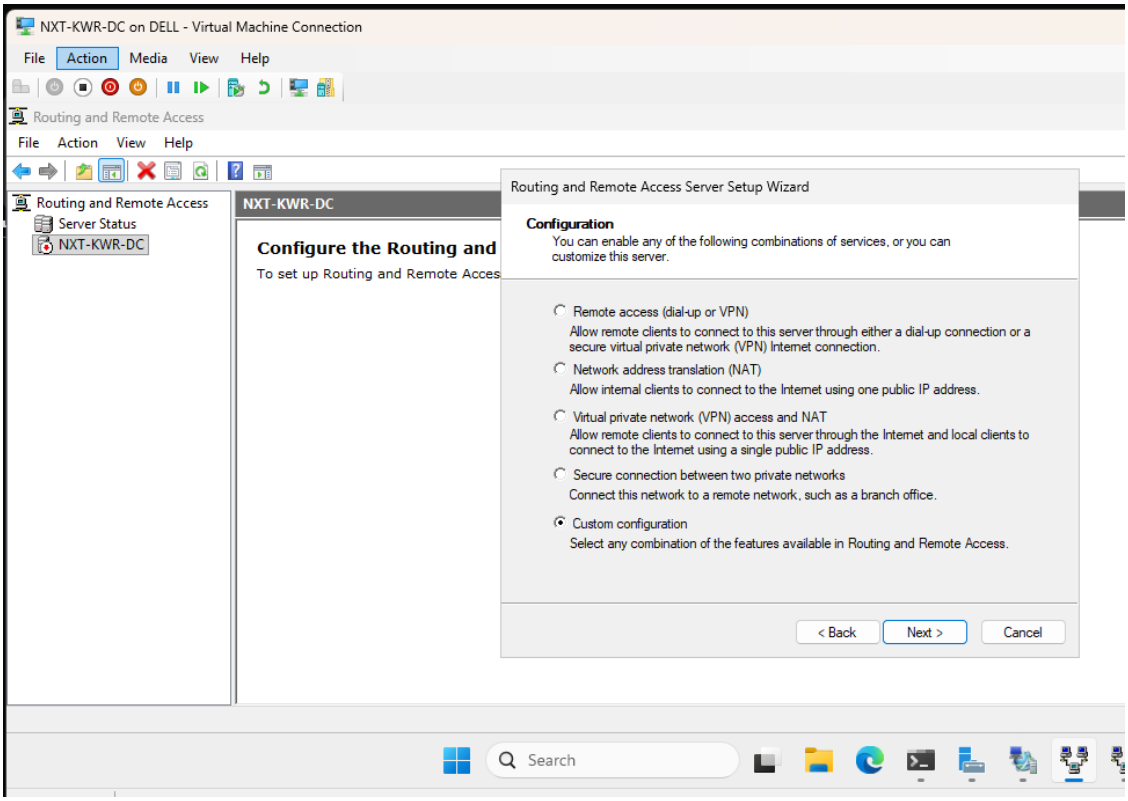


What this screen shows: The Routing and Remote Access MMC console, with the server node NXT-KWR-DC right-clicked, showing a context menu with Configure and Enable Routing and Remote Access at the top (Disable Routing and Remote Access is greyed out, since it isn't enabled yet), plus All Tasks, View, Delete, Refresh, Properties, and Help.

What to do here: Click Configure and Enable Routing and Remote Access.

What happens next: The Routing and Remote Access Server Setup Wizard launches, which is where the actual VPN/NAT configuration choice is made (next screenshot).

4.2.6 Choosing the RRAS configuration type

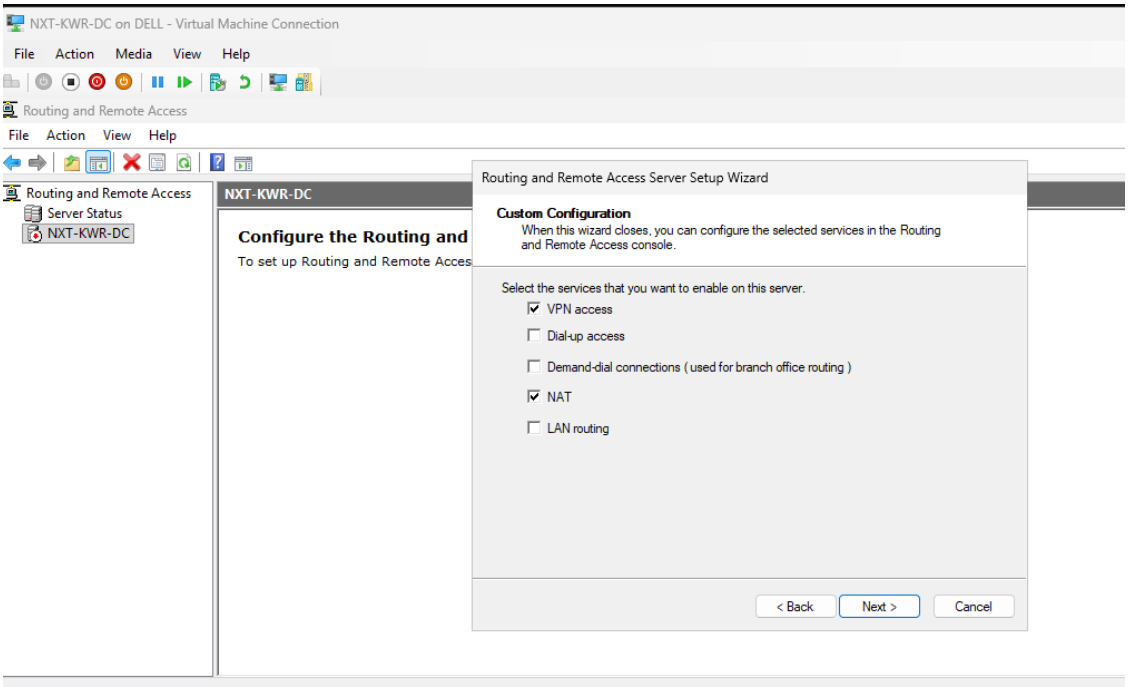


What this screen shows: The Routing and Remote Access Server Setup Wizard, Configuration page, offering five radio options: Remote access (dial-up or VPN); Network address translation (NAT); Virtual private network (VPN) access and NAT; Secure connection between two private networks; and Custom configuration (selected) — 'Select any combination of the features available in Routing and Remote Access.'

What to do here: Select Custom configuration rather than one of the pre-built combinations, because it gives explicit, individual control over exactly which services (VPN Access, NAT, etc.) are enabled — useful for understanding and troubleshooting the setup rather than accepting a bundled default. Click Next.

What happens next: The wizard shows a checklist of individual RRAS services to enable (next screenshot).

4.2.7 Selecting individual RRAS services



What this screen shows: The Custom Configuration page: a checklist with VPN access and NAT both ticked; Dial-up access, Demand-dial connections (used for branch office routing), and LAN routing all left unticked.

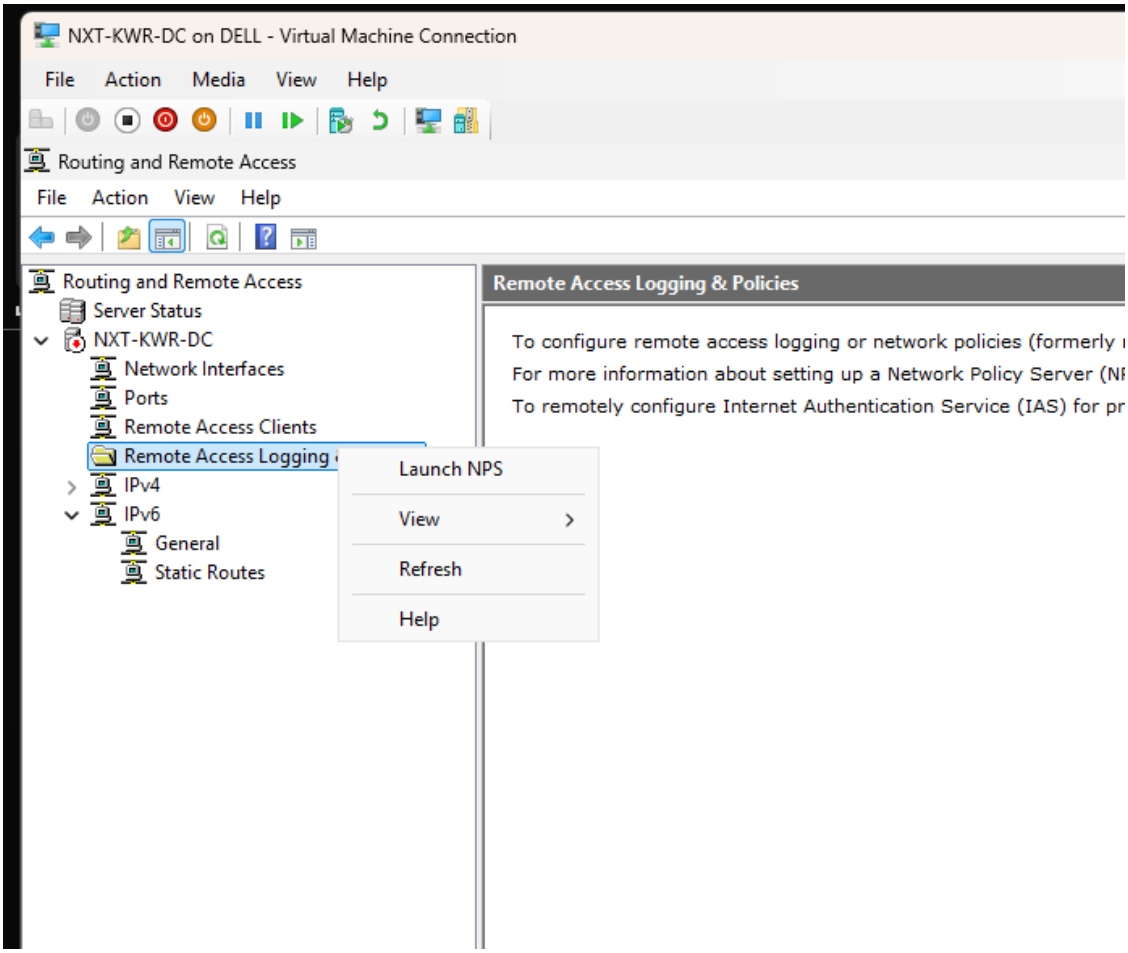
What to do here: Tick VPN access (to allow remote clients to connect via a VPN tunnel) and NAT (so those same clients — and/or internal clients — can reach the internet through this server's public interface). Leave Dial-up access unticked (legacy technology, not used here), Demand-dial connections unticked (that's for permanent site-to-site links, not needed for individual remote-access VPN), and LAN routing unticked (not routing between separate internal LAN segments in this scenario). Click Next, then Finish on the summary page.

What happens next: The wizard finishes gathering configuration and is ready to apply it — a confirmation prompt appears next, followed by the RRAS service actually starting.

After clicking Finish, a prompt appears confirming the configuration; click OK, and then start the Routing and Remote Access service if it does not start automatically. As a best practice, the notes recommend enabling Network Policy Server (NPS) network policies on this server next, covered below.

4.3 Configuring Network Policy Server (NPS) policies

4.3.1 Launching NPS from the RRAS console

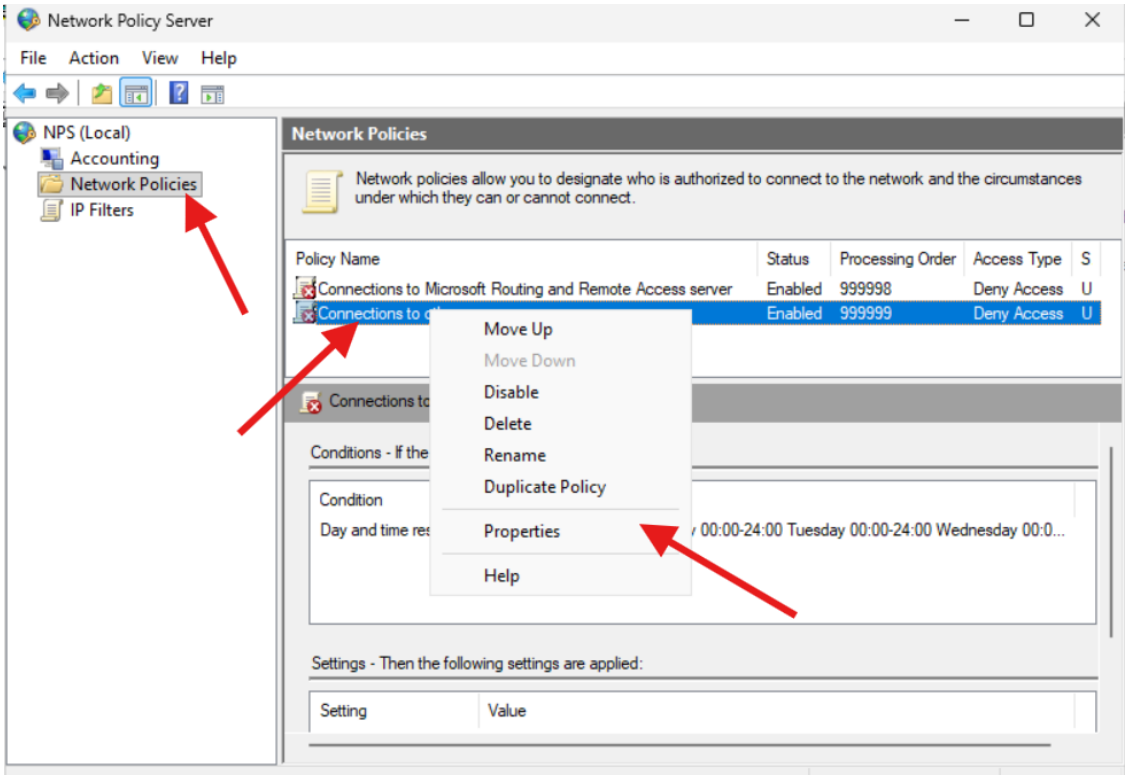


What this screen shows: The Routing and Remote Access console, expanded to show Server Status, NXT-KWR-DC (with Network Interfaces, Ports, Remote Access Clients, and Remote Access Logging & Policies sub-nodes, plus IPv4/IPv6). Remote Access Logging & Policies is right-clicked, showing a context menu with Launch NPS at the top, plus View, Refresh, and Help. The right-hand pane explains that to configure remote access logging or network policies (formerly IAS), you can remotely configure Internet Authentication Service (IAS) for more information.

What to do here: Click Launch NPS.

What happens next: The separate Network Policy Server (NPS) management console opens, where the actual access-control policies for incoming VPN connections are configured.

4.3.2 Reviewing and reordering network policies

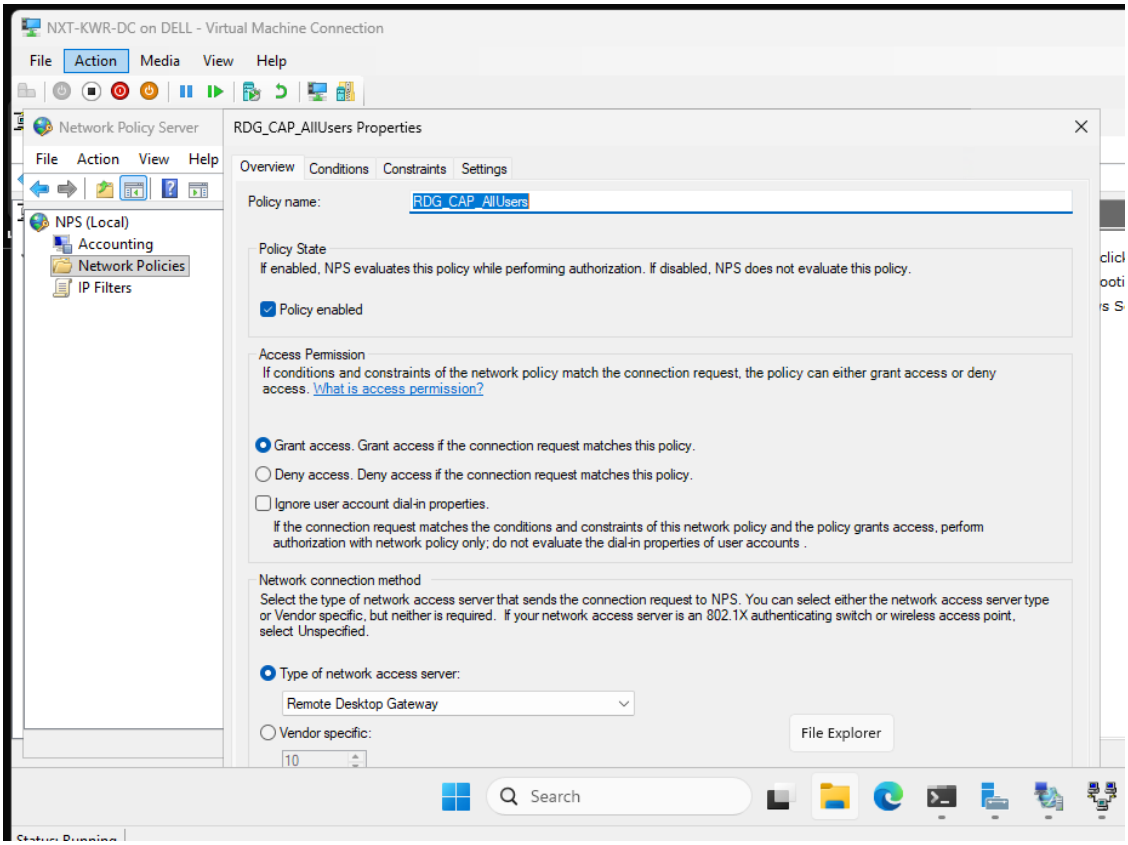


What this screen shows: The Network Policy Server console, Policies → Network Policies node. Two default policies are listed: 'Connections to Microsoft Routing and Remote Access server' (Enabled, Processing Order 999998, Access Type Deny Access) and 'Connections to...' (Enabled, Processing Order 999999, Access Type Deny Access, currently selected/highlighted with a right-click context menu open showing Move Up, Move Down (greyed out), Disable, Delete, Rename, Duplicate Policy, Properties, and Help). Two red arrows point at the selected policy and at the Properties menu item, and a Conditions panel below shows a Day-and-time-restrictions condition allowing Monday/Tuesday/Wednesday 00:00–24:00.

What to do here: Both default built-in policies deny access by default — this is a safety default so that no VPN connection is accidentally allowed before an explicit policy grants it. Right-click the relevant policy and click Properties to change its behaviour to Grant access.

What happens next: The policy's Properties dialog opens on the Overview tab, where Grant access can actually be selected (next screenshot).

4.3.3 Granting access on a network policy

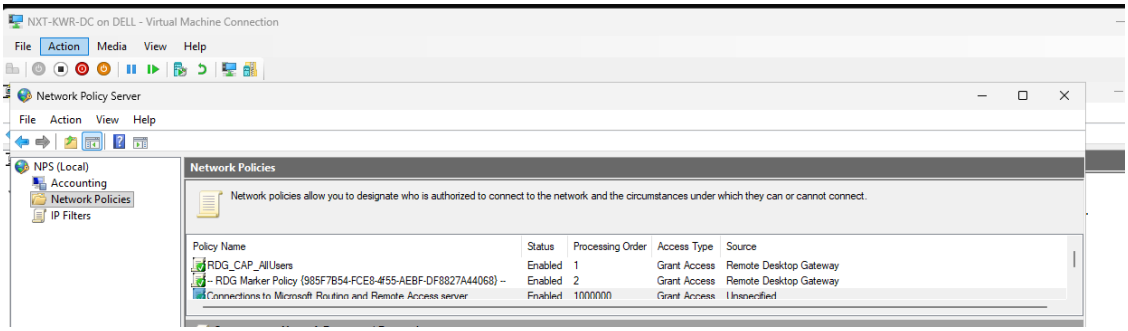


What this screen shows: The RDG_CAP_AllUsers Properties dialog, Overview tab: Policy Enabled is ticked; under Access Permission, Grant access — 'Grant access if the connection request matches the policy' — is selected (rather than Deny access); Ignore user account dial-in properties is unticked; Network connection method is set to Type of network access server = Remote Desktop Gateway.

What to do here: Select Grant access (instead of the default Deny access) so that connections matching this policy's conditions are actually allowed through, then click Apply and OK, as instructed in the source notes: 'then click Grant access and click apply and ok'.

What happens next: Any connection attempt that matches this policy's conditions (for example, being a member of the right group, connecting within the allowed time window) will now be permitted instead of automatically denied.

4.3.4 Confirming the updated policy list

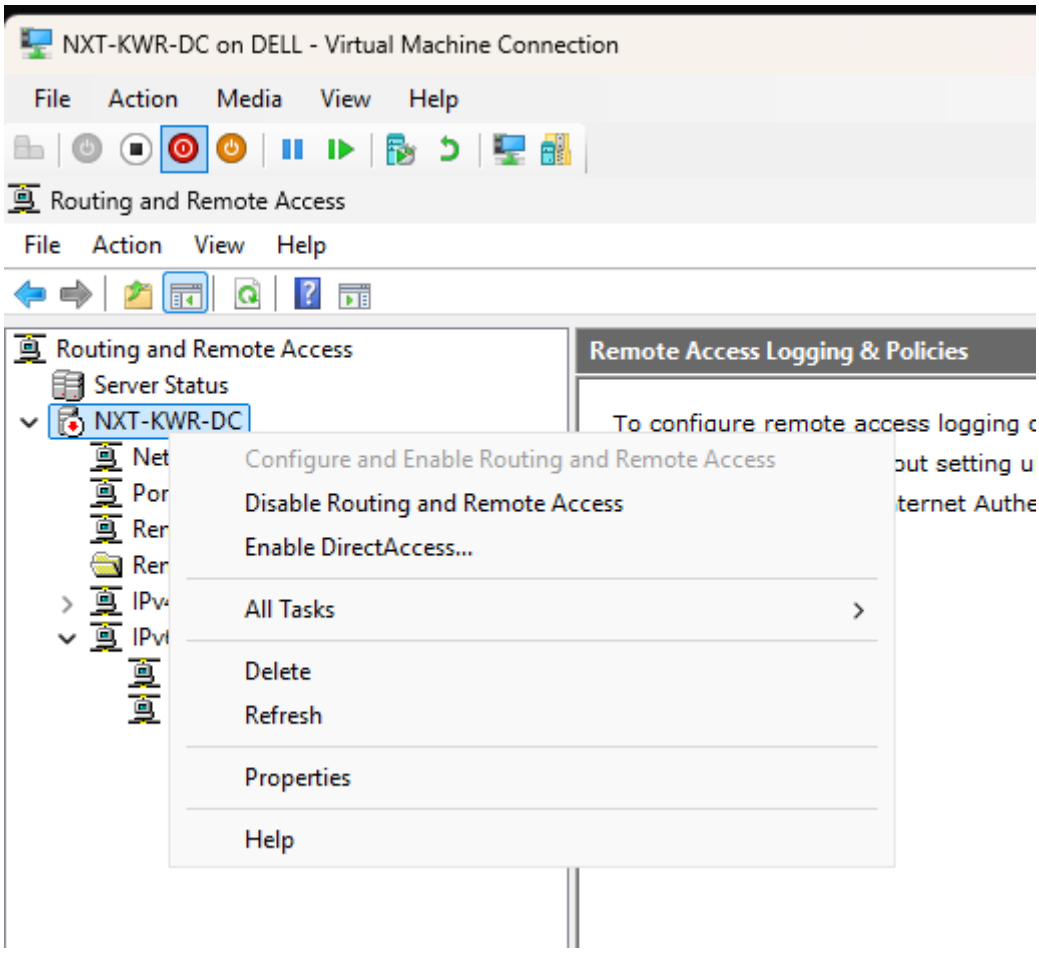


What this screen shows: The Network Policies list now showing three policies: RDG_CAP_AllUsers (Status Enabled 1, Access Type Grant Access, Source Remote Desktop Gateway), RDG Marker Policy (9E9F7804-FCE9-4131-4E8F-DF9827A4406E) (Status Enabled 2, Grant Access, Remote Desktop Gateway), and Connections to Microsoft Routing and Remote Access server (Status Enabled 999999, Access Type Unspecified).

What to do here: Confirm the policy that was just edited now shows Grant Access rather than Deny Access.

What happens next: Incoming VPN/Gateway connection attempts will now be evaluated against these policies in processing-order sequence, and users who match the Grant Access policy will be allowed through instead of rejected.

4.3.5 Verifying the RRAS service is running



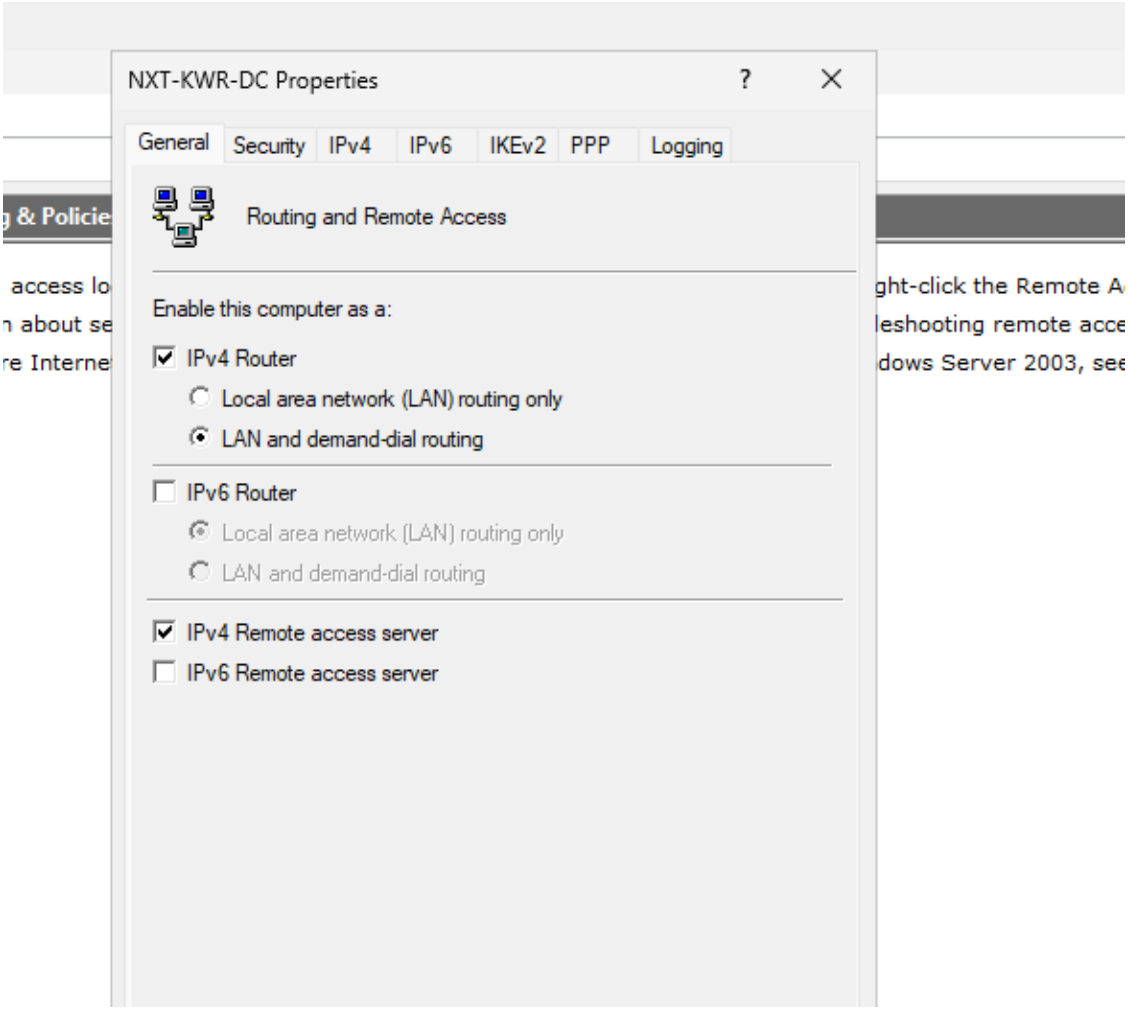
What this screen shows: Back in the Routing and Remote Access console, NXT-KWR-DC right-clicked, showing a context menu where Configure and Enable Routing and Remote Access and Disable Routing and Remote Access are both greyed out (because it is already enabled and running), but Enable DirectAccess..., All Tasks, Delete, Refresh, Properties, and Help remain available.

What to do here: This confirms the RRAS service is already configured and running (the greyed-out Configure/Disable options are the visual cue) — no action is needed here beyond verifying the state.

What happens next: The server is now actively running as a VPN/NAT gateway; the remaining steps configure the address pool clients will draw from and the actual NAT rule on the internet-facing interface.

4.4 Configuring the VPN address pool

4.4.1 Enabling the server as an IPv4 router

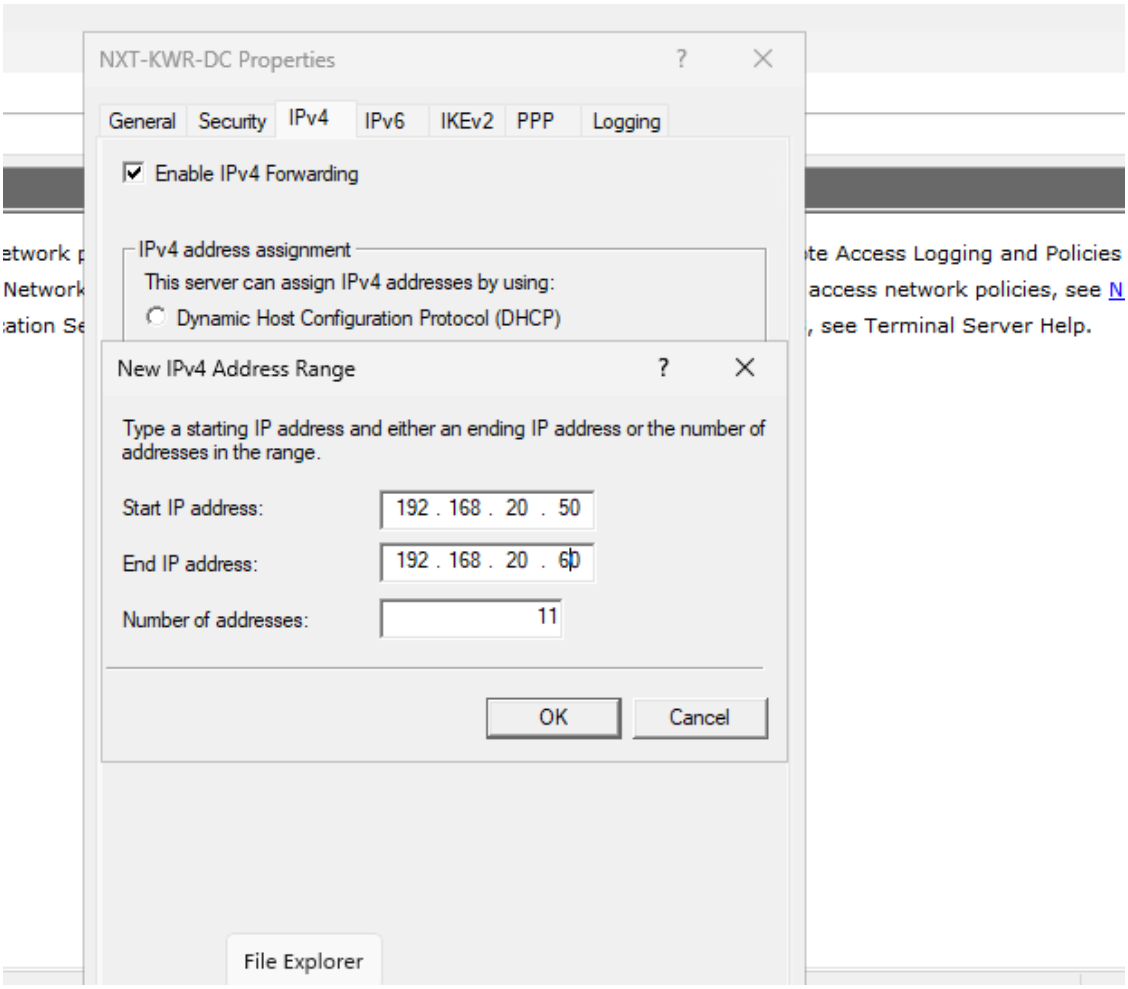


What this screen shows: NXT-KWR-DC Properties, General tab, listing checkboxes: IPv4 Router (ticked) with sub-options Local area network (LAN) routing only vs LAN and demand-dial routing (LAN and demand-dial routing selected), IPv6 Router (unticked), IPv4 Remote access server (ticked), IPv6 Remote access server (unticked).

What to do here: Confirm IPv4 Router is enabled with LAN and demand-dial routing selected, and IPv4 Remote access server is ticked (IPv6 is left disabled here since this walkthrough only uses IPv4 addressing).

What happens next: The server is now explicitly permitted to both route IPv4 traffic between interfaces and accept incoming IPv4 remote-access (VPN) connections — the prerequisite for the address-pool configuration on the next tab.

4.4.2 Setting a static IPv4 address range for VPN clients



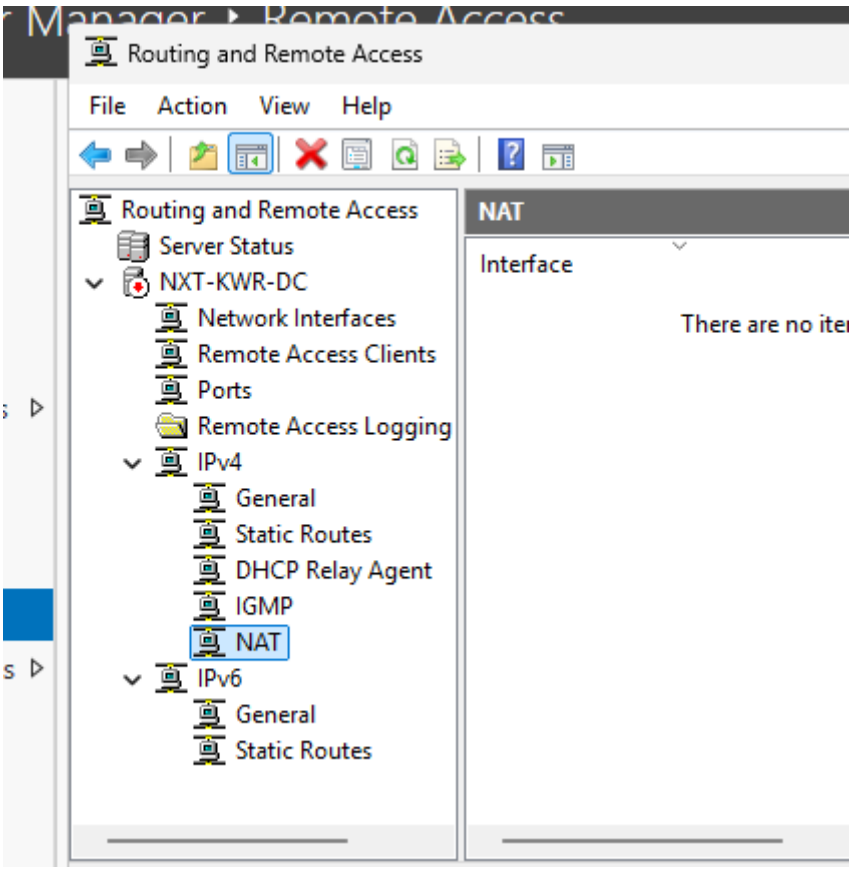
What this screen shows: The same Properties dialog, IPv4 tab: Enable IPv4 Forwarding is ticked; under IPv4 address assignment, 'This server can assign IPv4 addresses by using' offers Dynamic Host Configuration Protocol (DHCP) or Static address pool (Static address pool selected). A New IPv4 Address Range dialog is open on top, with Start IP address 192.168.20.50, End IP address 192.168.20.60, and Number of addresses calculated as 11.

What to do here: Choose Static address pool (rather than relying on an external DHCP server to hand out addresses to VPN clients) and click Add to define a range — here 192.168.20.50 through 192.168.20.60, giving 11 usable addresses for simultaneous VPN clients — then click OK.

What happens next: Every client that successfully connects over VPN will now be handed an address from this dedicated 192.168.20.50–.60 pool, keeping VPN clients clearly identifiable and separate from normal DHCP-assigned LAN clients.

4.5 Configuring NAT on the internet-facing interface

4.5.1 Opening the NAT node

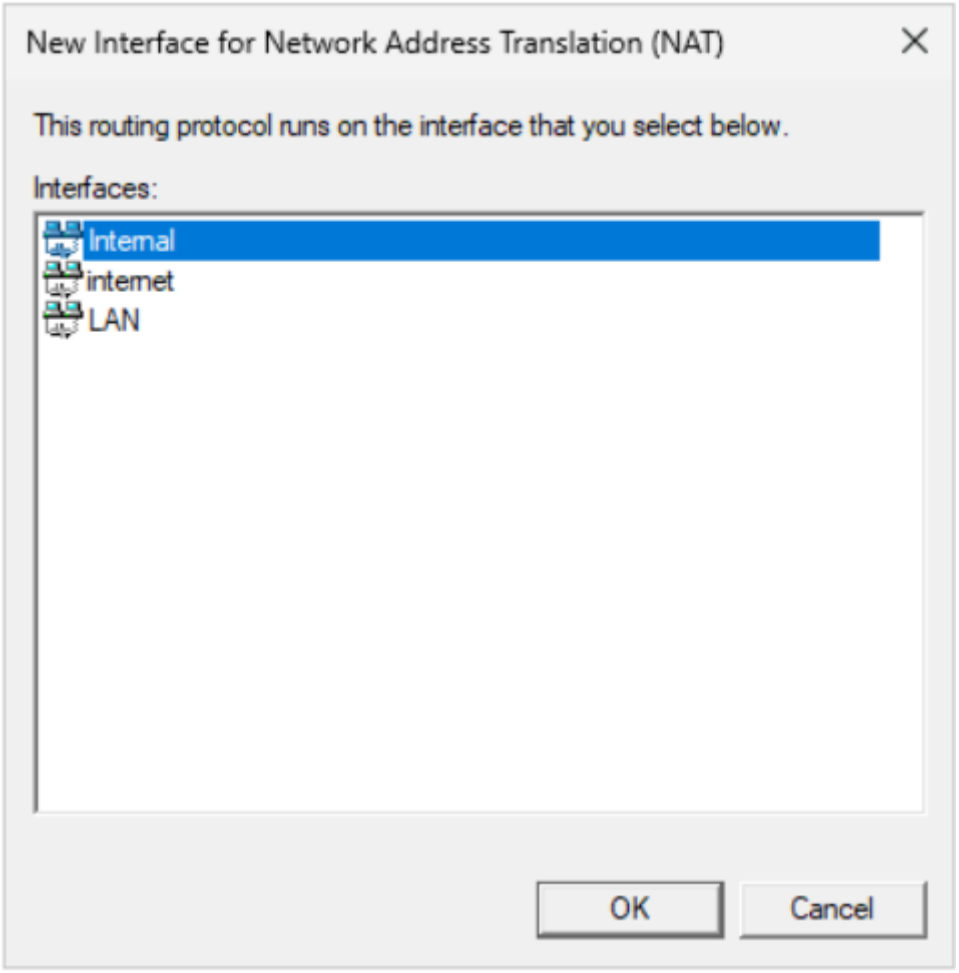


What this screen shows: The Routing and Remote Access console tree, IPv4 expanded to show General, Static Routes, DHCP Relay Agent, IGMP, and NAT (selected). The right-hand pane, titled NAT, currently reads 'There are no items to show in this view.'

What to do here: Right-click the NAT node and choose New Interface... to add the interface that NAT should apply to.

What happens next: The New Interface for Network Address Translation dialog opens, listing the server's available interfaces to choose from.

4.5.2 Choosing which interface gets NAT

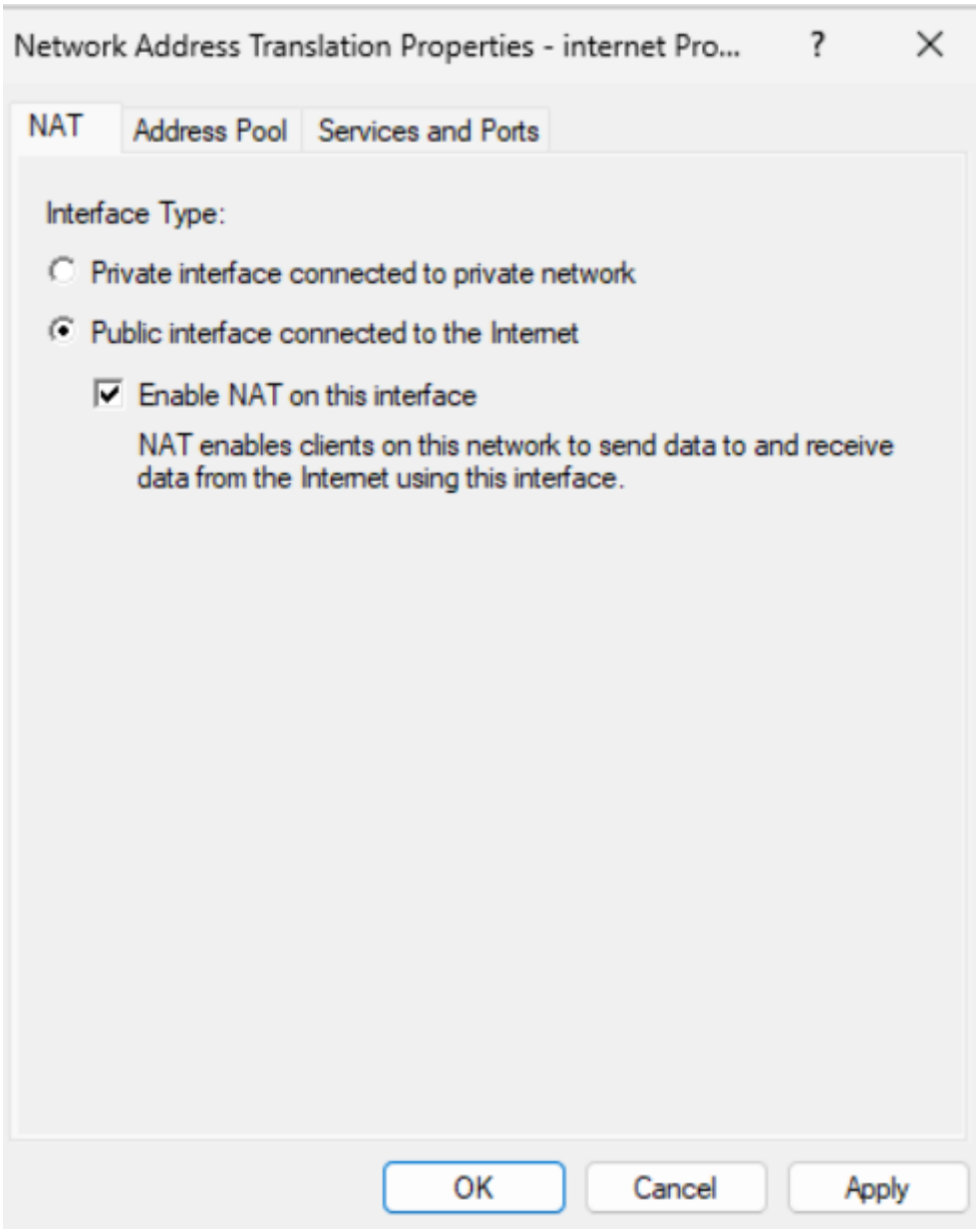


What this screen shows: The New Interface for Network Address Translation dialog, listing three interfaces: Internal, internet (highlighted/selected), and LAN.

What to do here: Select the interface that faces the internet — named 'internet' in this environment (this is the second network adapter configured back in step 4.1.2, with the 172.20.10.x address and default gateway) — and click OK. This is deliberately not the internal/LAN interface, since NAT needs to translate traffic on the interface that actually leads outward.

What happens next: The Network Address Translation Properties dialog opens for that interface, where its public/private role and NAT behaviour are explicitly configured (next screenshot).

4.5.3 Marking the interface as public and enabling NAT



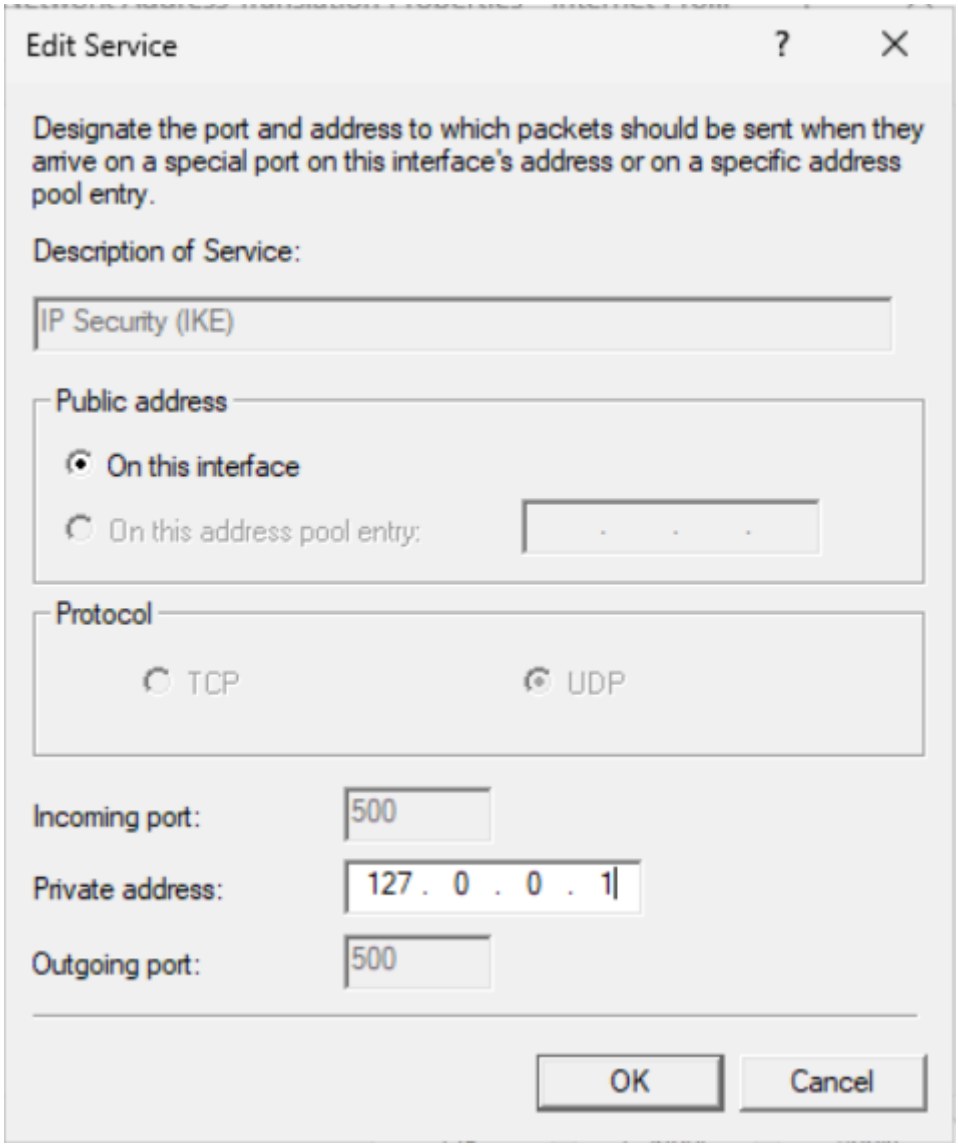
What this screen shows: The Network Address Translation Properties — internet dialog, NAT tab. Two Interface Type radio options: Private interface connected to private network, and Public interface connected to the Internet (selected). Below, a checkbox 'Enable NAT on this interface' is ticked, with explanatory text that NAT enables clients on this network to send data to and receive data from the Internet using this interface.

What to do here: Select Public interface connected to the Internet (matching the earlier guidance that a private interface carries 192.168.x.x/172.16–31.x.x internal addressing, while a public interface faces the internet) and tick Enable NAT on this interface, then click OK.

What happens next: Traffic originating from internal clients (and from VPN clients using the pool configured earlier) can now be translated through this interface's public IP address to reach the internet, exactly the way a home router NATs an entire household's internal devices behind one public IP.

4.6 Configuring the IKE (IPSec) service for VPN traffic

4.6.1 Editing the IP Security (IKE) service

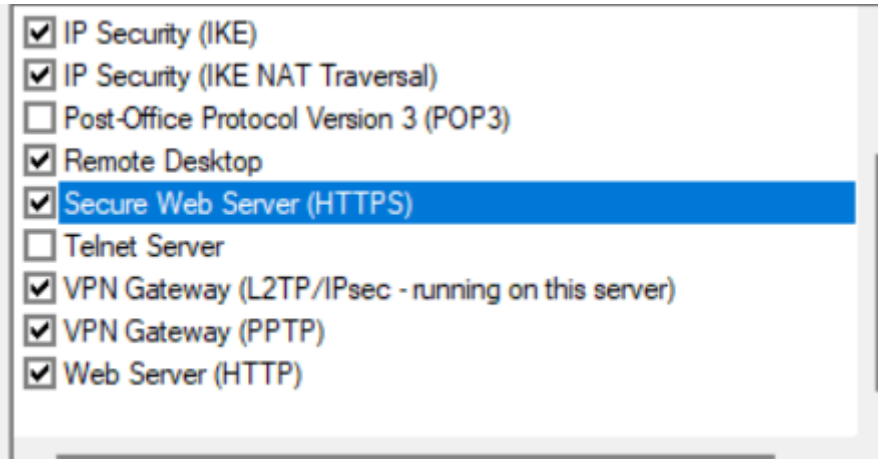


What this screen shows: The Edit Service dialog, opened from the NAT interface's Services and Ports tab, for the IP Security (IKE) service. Fields: Public address — On this interface (selected); Protocol — UDP (selected, TCP unavailable for this service); Incoming port 500; Private address 127.0.0.1 (highlighted); Outgoing port 500.

What to do here: With Public address set to 'On this interface' and Protocol UDP already fixed by the service definition, set Private address to 127.0.0.1, since — as the accompanying notes explain — this refers to the local server itself (the RRAS server is both the public-facing endpoint and the actual VPN endpoint terminating the IKE negotiation, so the traffic is redirected back to the server itself rather than to a separate internal host). Click OK.

What happens next: Incoming UDP port 500 traffic arriving on the public/internet interface (used for IKE, the key-exchange protocol behind IPSec-based VPN types such as IKEv2) is now correctly forwarded to the RRAS server itself, allowing IPSec/IKE-based VPN negotiation to succeed through NAT.

4.6.2 Enabling the remaining VPN-related services and ports

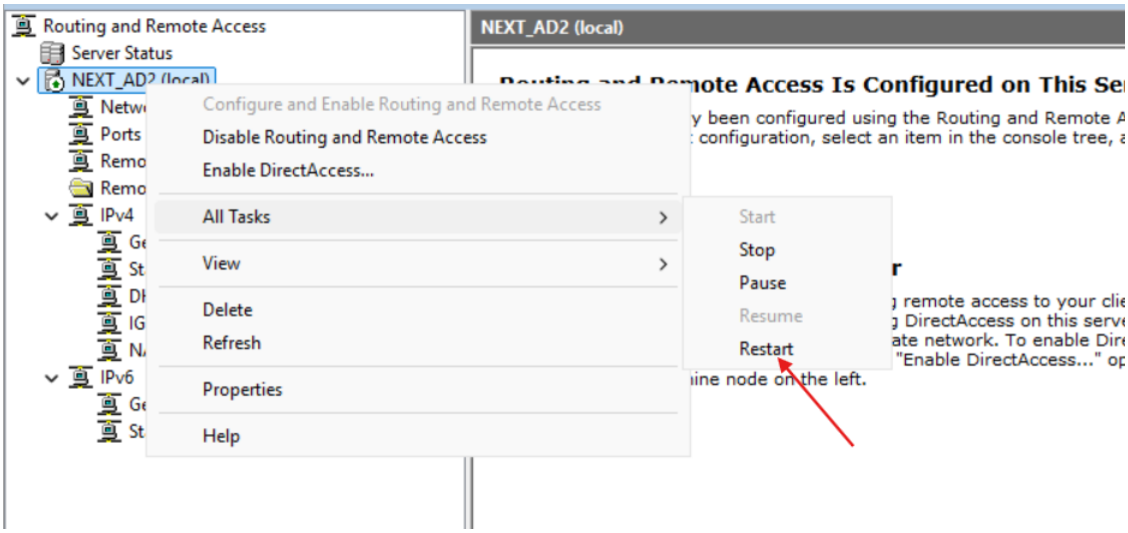


What this screen shows: The Services and Ports checklist for the NAT interface, showing several services ticked: IP Security (IKE), IP Security (IKE NAT Traversal), Remote Desktop, Secure Web Server (HTTPS) (highlighted/selected), VPN Gateway (L2TP/IPsec - running on the server), VPN Gateway (PPTP), and Web Server (HTTP); Post-Office Protocol Version 3 (POP3) and Telnet Server are left unticked.

What to do here: Tick every VPN-relevant service the same way IP Security (IKE) was just configured — IP Security (IKE NAT Traversal), VPN Gateway (L2TP/IPsec), and VPN Gateway (PPTP) — so that whichever protocol a given client uses (IKEv2/IPsec, L2TP/IPsec, or the older PPTP) can reach through NAT to the server. Secure Web Server (HTTPS) and Web Server (HTTP) are also ticked here since RD Web Access/RD Gateway (from Part 3) share this same server and need their ports open too. Leave POP3 and Telnet unticked — they are unrelated legacy services not used in this deployment.

What happens next: Each ticked entry opens (forwards) the corresponding port through the NAT interface to the server itself, following the same 'apply local address to private address (127.0.0.1)' pattern used for IKE — without this, a VPN client's connection attempt on that protocol's port would be silently dropped at the NAT boundary.

4.6.3 Restarting the server to finalise the configuration



What this screen shows: The Routing and Remote Access console, NXT-KWR-DC (local) node right-clicked, showing All Tasks expanded with Start, Stop, Pause, Resume (greyed out), and Restart — Restart is indicated with a red arrow. The right-hand pane heading reads 'Routing and Remote Access Is Configured on This Server', with body text (partially visible) about remote access having been configured using the Routing and Remote Access wizard and network settings.

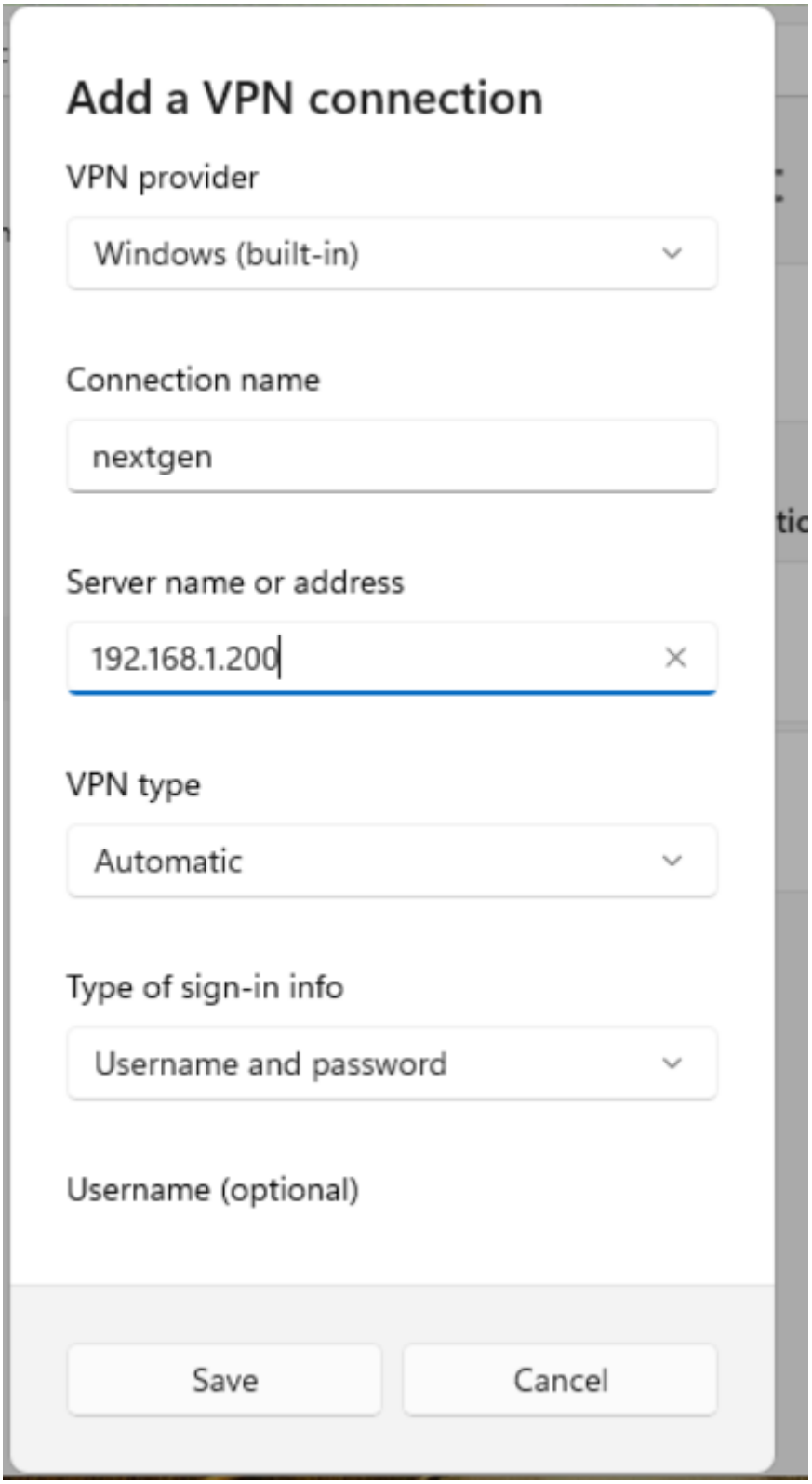
What to do here: Click All Tasks → Restart to restart the Routing and Remote Access service so that the NAT interface, address pool, and service/port changes just made all take effect together.

What happens next: The RRAS service restarts and comes back up with the complete configuration active — VPN Access enabled, NAT enabled and bound to the internet interface, the static address pool ready to hand out to connecting clients, and every relevant VPN port opened through NAT.

Part 5 — Connecting a Client to the VPN

With the server side fully configured, the final steps are to create/authorise an Active Directory user for VPN dial-in (the same Dial-in tab used back in Part 1.3 for RDS) and then build the actual VPN connection profile on the client machine.

5.1 Creating the VPN connection profile on the client



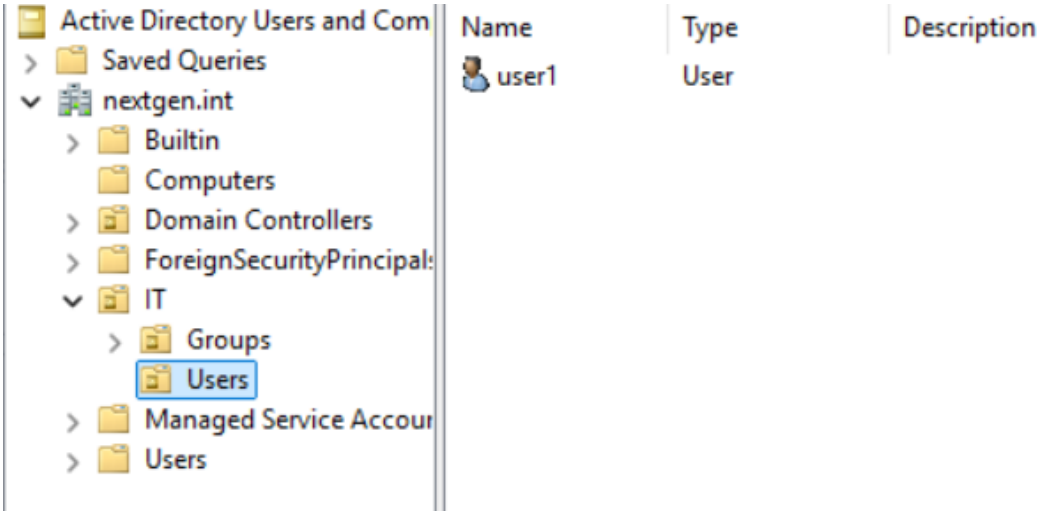
What this screen shows: The Windows 'Add a VPN connection' settings page: VPN provider = Windows (built-in); Connection name = nextgen; Server name or address = 192.168.1.200; VPN type = Automatic; Type of sign-in info = Username and password; Username (optional) field empty; Save/Cancel buttons.

What to do here: Open Settings → Network & Internet → VPN → Add a VPN connection on the client PC. Set VPN provider to Windows (built-in), give the connection a recognisable name (nextgen), and set Server name or address to the public/internet-facing IP address of the RRAS server (192.168.1.200 in this lab, i.e. the address the client can actually reach over the network it's on). Leave VPN type as Automatic so Windows negotiates the best available protocol (it will typically try IKEv2 first, matching the IKE service just opened on the server), select Username and password as the sign-in method, and click Save.

What happens next: A new VPN connection entry named 'nextgen' appears in the client's network list, ready to be connected — but it will only succeed once the Active Directory user attempting to sign in has both dial-in permission and the correct group membership, configured next.

5.2 Authorising the Active Directory user for VPN dial-in

5.2.1 Locating the user account

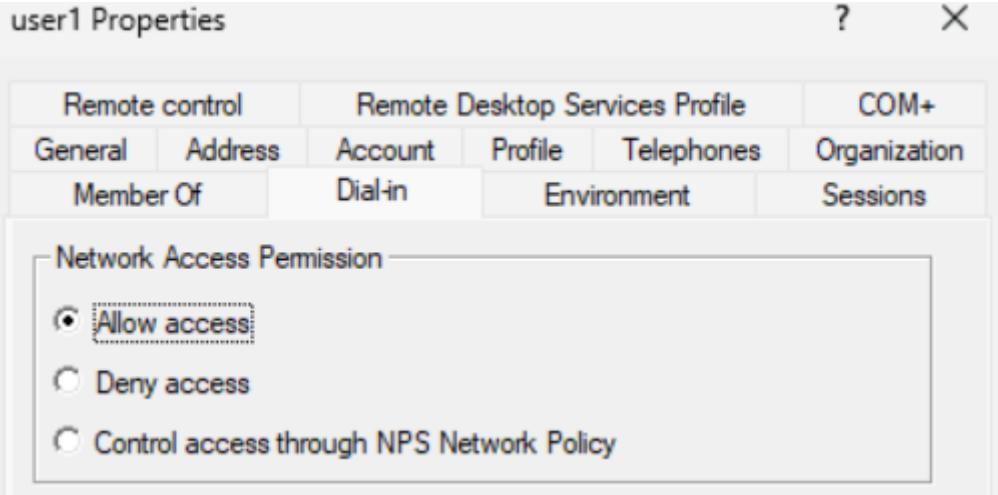


What this screen shows: Active Directory Users and Computers, showing the nextgen.int domain tree expanded to IT → Users, with a single account named user1 (Type: User) listed in the right-hand pane.

What to do here: Right-click user1 and open Properties, exactly as was done for the RDS test user back in Part 1.3.

What happens next: The familiar multi-tab Properties dialog opens for this account.

5.2.2 Allowing network access on the Dial-in tab

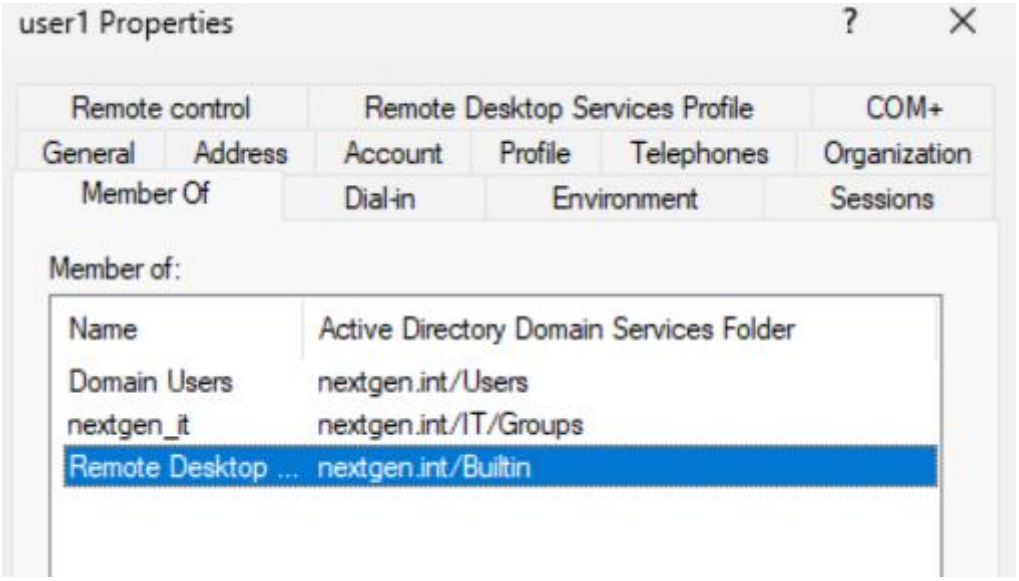


What this screen shows: user1 Properties, Dial-in tab, Network Access Permission section with three radio buttons: Allow access (selected), Deny access, Control access through NPS Network Policy.

What to do here: Select Allow access (the same setting used in Part 1.3.1 for RDS access) — this is the account-level switch RRAS checks before even considering the NPS network policy conditions configured back in Part 4.3.

What happens next: The account is now individually authorised to establish a remote-access (dial-in/VPN) connection at all; without this, the connection attempt is rejected before NPS policy evaluation even happens.

5.2.3 Confirming group membership for VPN access



What this screen shows: user1 Properties, Member Of tab, listing three groups: Domain Users (nextgen.int/Users), nextgen_it (nextgen.int/IT/Groups), and Remote Desktop... (nextgen.int/Builtin, selected/highlighted).

What to do here: As the source notes state, the user must be in the remote access group as well as having dial-in access allowed — confirm the account belongs to whichever group the NPS network policy's conditions reference (here shown alongside Remote Desktop Users and the nextgen_it group) so the Grant Access policy from Part 4.3.3 actually matches this account. Click Apply and OK.

What happens next: With both the account-level Allow access permission and the correct group membership in place, user1 can now click Connect on the 'nextgen' VPN profile created in step 5.1, authenticate with their domain username and password, be assigned an address from the 192.168.20.50–.60 pool, and reach authorised internal resources (including the RDS deployment from Parts 1–3) through the encrypted VPN tunnel — completing the deployment end-to-end.

Summary

- Part 1: Authorised a test user for Remote Desktop and installed the RDS role (Connection Broker, Web Access, Session Host) via Quick Start / session-based deployment.
- Part 2: Covered VPN theory — why VPNs are used, the taxonomy of VPN types, the underlying protocols (IPSec, SSL/TLS, L2TP, OpenVPN, WireGuard), and security best practices.
- Part 3: Installed RD Licensing (Per Device/Per User CALs) and RD Gateway, created and applied a self-signed certificate across every role service, published RemoteApp programs (Calculator, Notepad, Paint, XPS Viewer), and verified end-to-end access from both a domain-joined admin session and an external client with the certificate imported.
- Part 4: Installed the Remote Access (RRAS) role, chose Deploy VPN only with a Custom configuration enabling VPN access and NAT, configured a static VPN client address pool, bound NAT to the internet-facing interface, and opened the IKE/L2TP/PPTP/HTTPS ports needed for VPN clients to reach the server through NAT.
- Part 5: Created the client-side VPN connection profile and authorised the Active Directory user for dial-in access and group membership, completing a working remote-access VPN into the nextgen.int network.